



AUBURN UNIVERSITY

SAMUEL GINN
COLLEGE OF ENGINEERING

3D Modeling of Tin Whiskers

Group 2

ORR

MANAGER: Cameron Brock

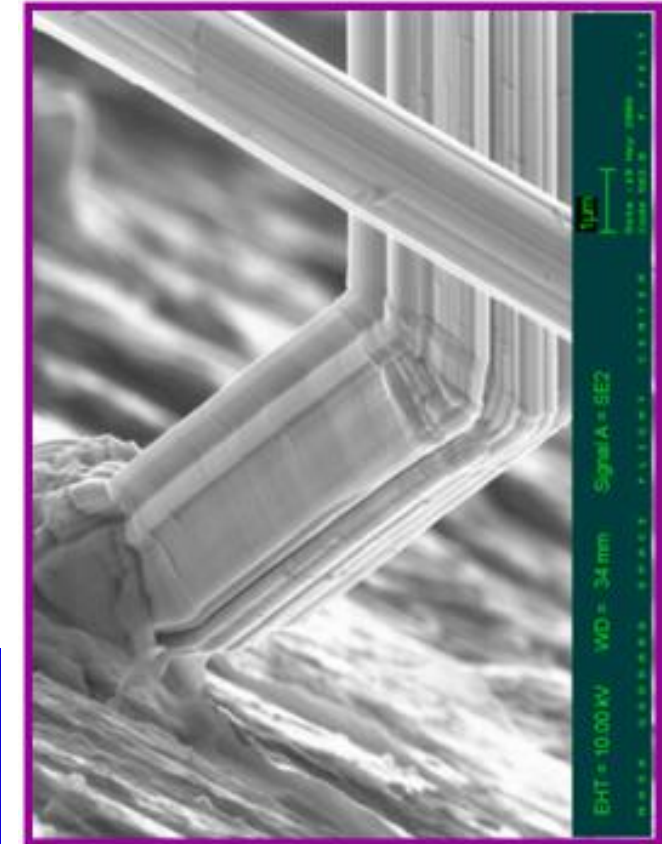
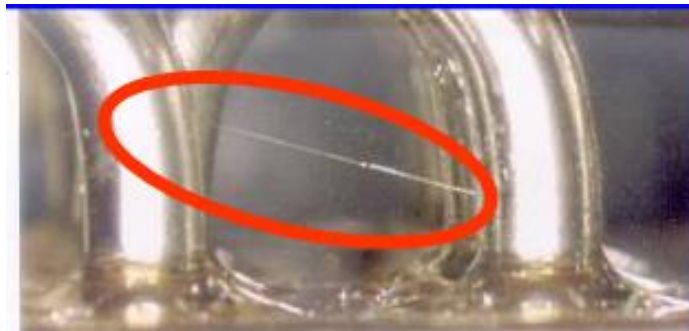
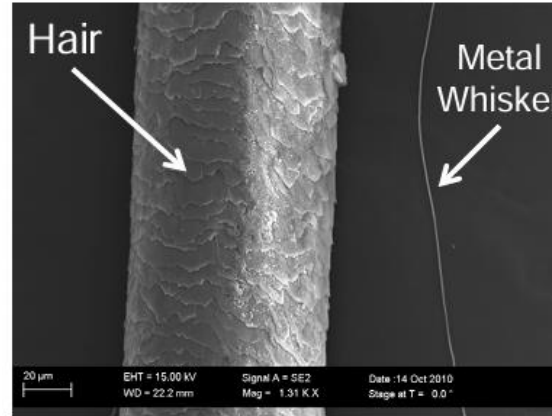
SCRIBE: Austin Hill

Roy Chen, Daniel Wheeler

What are Tin Whiskers?

- Microscopic, hair-like metal structures that grow outward from a grain or several grains on a metal surface
- Whiskers most commonly form on tin, zinc, and cadmium coatings
 - Also been observed on metals like indium, gold, silver, and lead
- Atoms from the metal itself migrate along grain boundaries and accumulate at the whisker's base (not the tip), causing growth
- Scientists STILL cannot reliably predict when, how many, or how large whiskers will grow

SEM comparison of
Human Hair vs. Metal Whisker



*Tin Whiskers on Tin-Plated
Electromagnetic Relay Terminals*

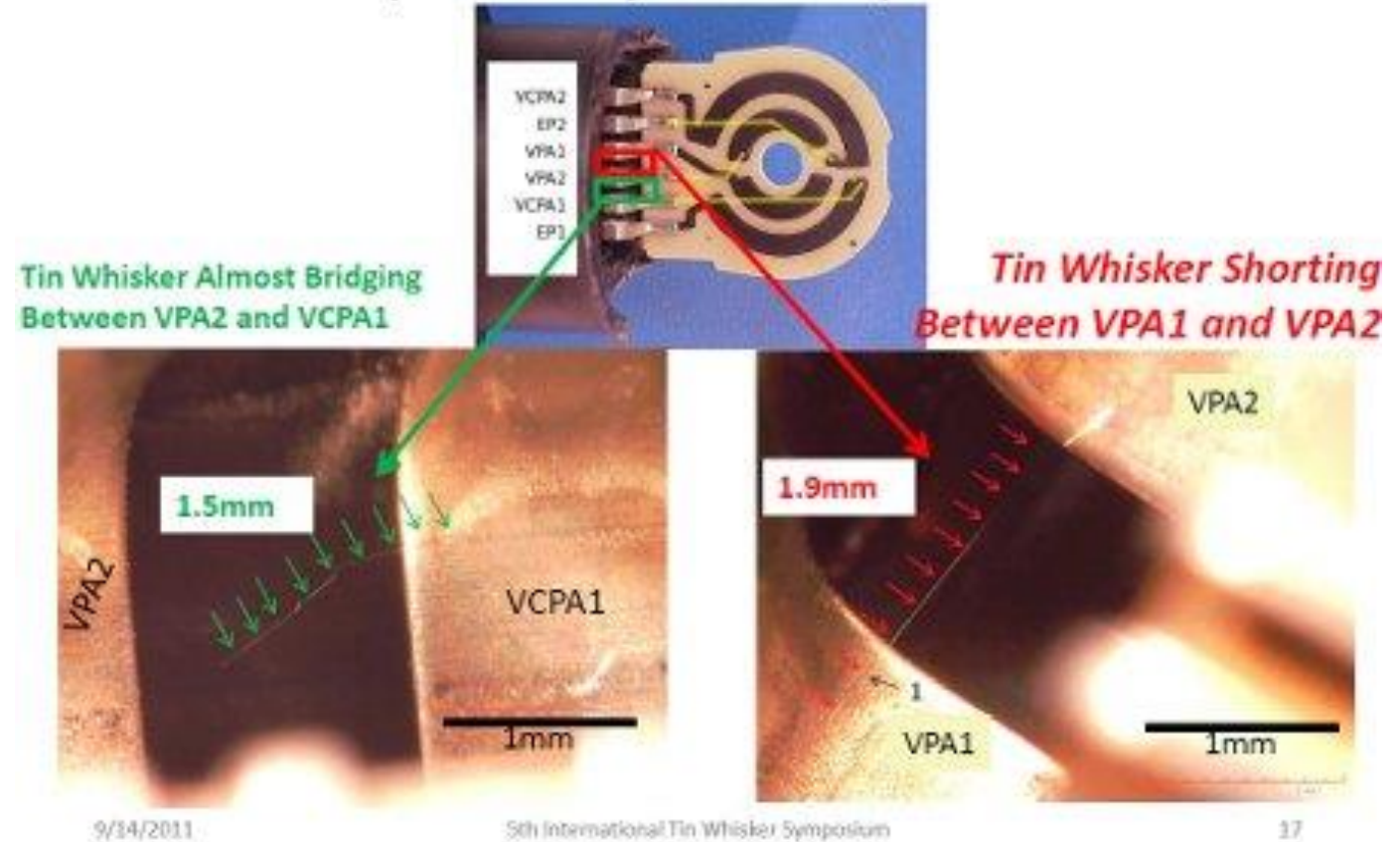
Problem Statement

- Printed Circuit Boards (PCB) are within every device with an electronic function
- Tin whiskers spontaneously grown on metal surfaces within PCBs and cause electrical short circuits
- Whiskers have caused failures in satellites, relay terminals, missile systems, and medical devices
- Current mitigation strategies remain inconsistent



- Starting around 2003, Toyota used tin-plated terminals in their electronic throttle control systems (ETCS)
- Reports of unintended increases in acceleration in several Toyotas
- Investigations suggested that tin whisker growth in the ETCS may have contributed to shorts
- Multiple accidents were linked to sudden speed increases

The Two Longest Tin Whiskers Observed in Faulty 2003 Toyota Camry APP Sensor

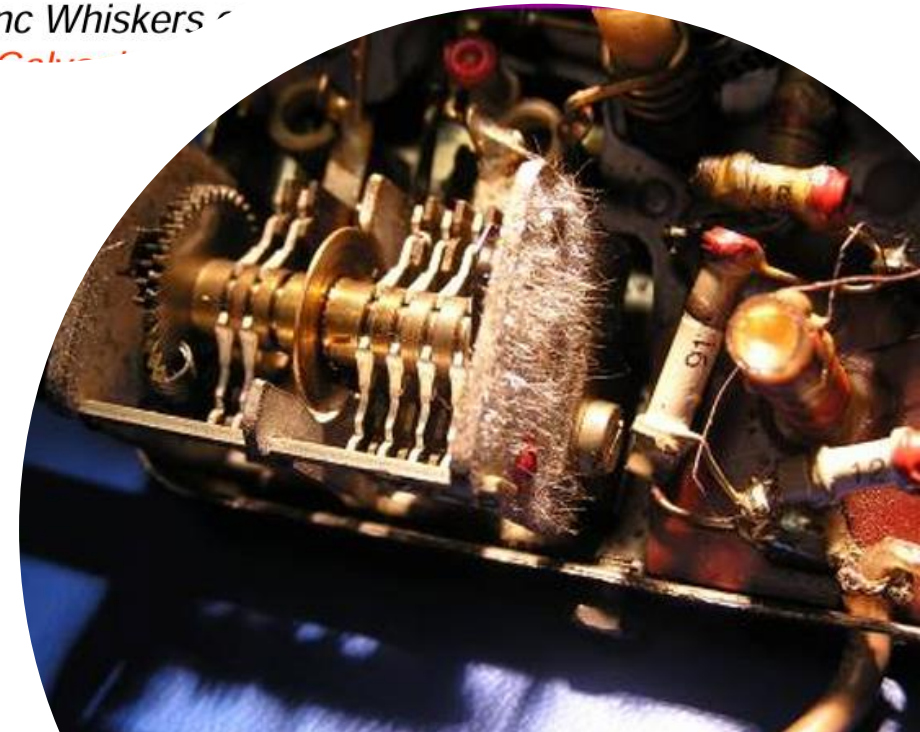


Objective

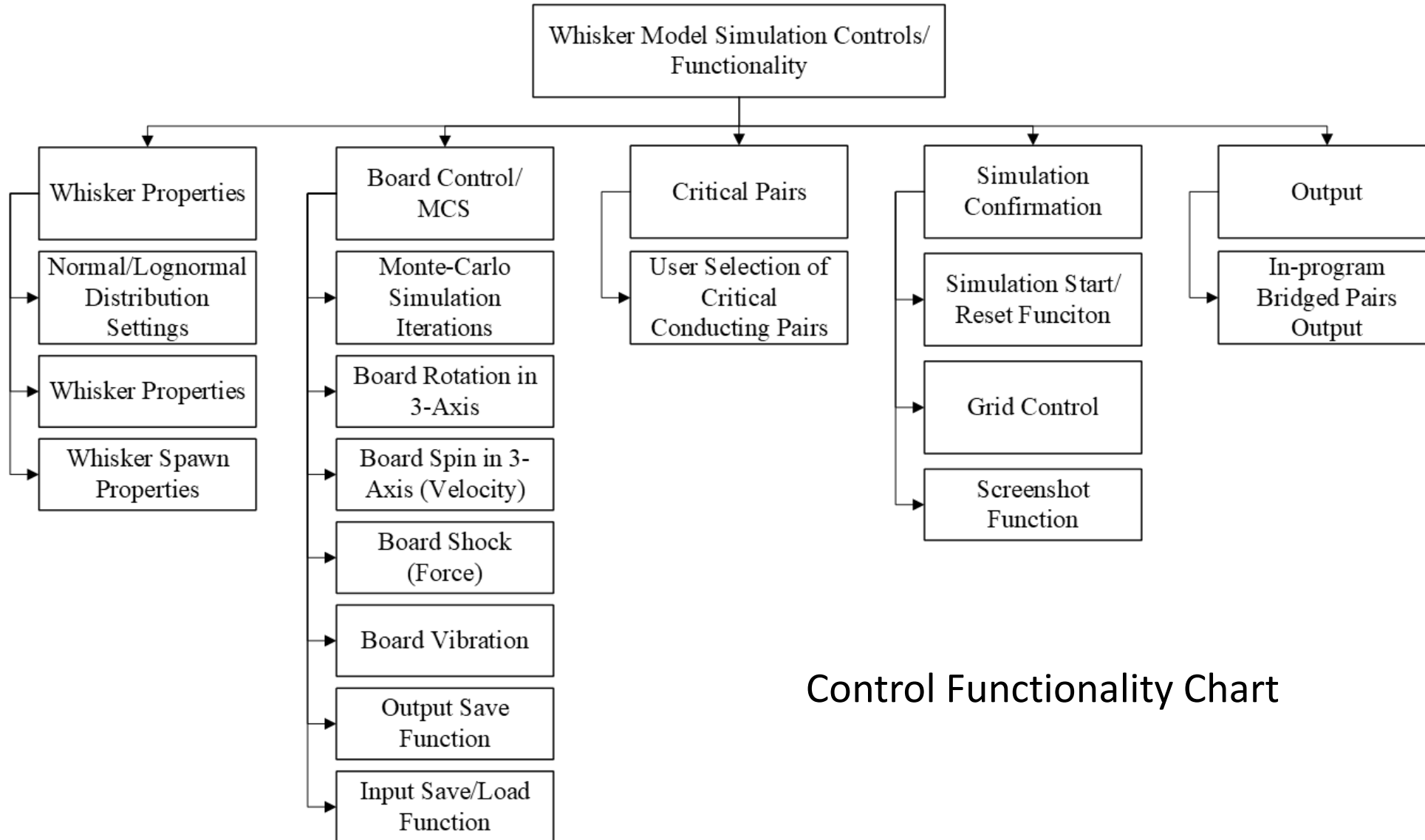
- Create a useful predictive simulation tool
 - Unity game engine
 - Monte Carlo techniques
- Improve previous group's tool
 - Model effects of electrostatic forces
 - Upgrade usability with PCB import
 - Model external forces



Zinc Whiskers



Background Information

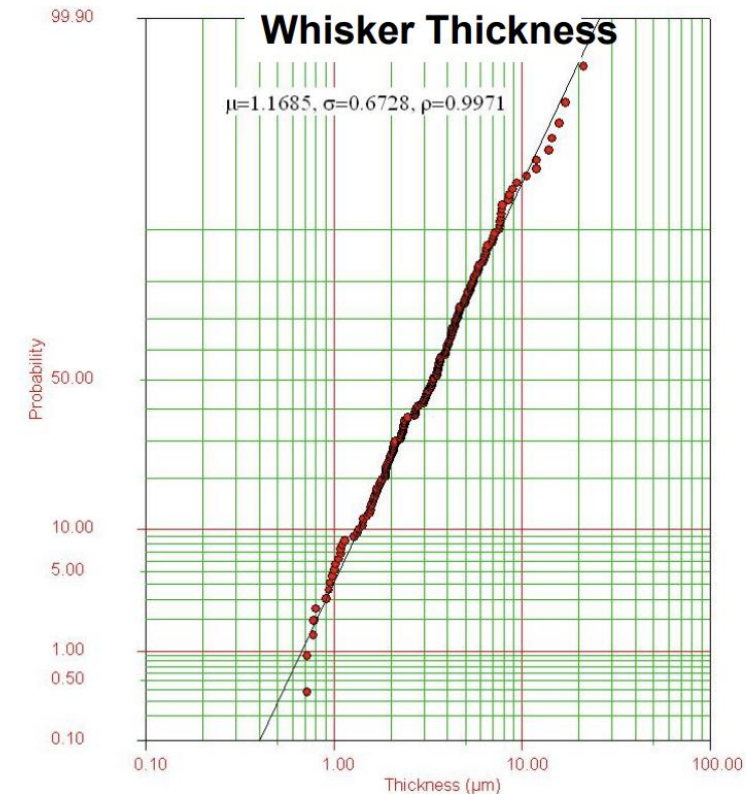
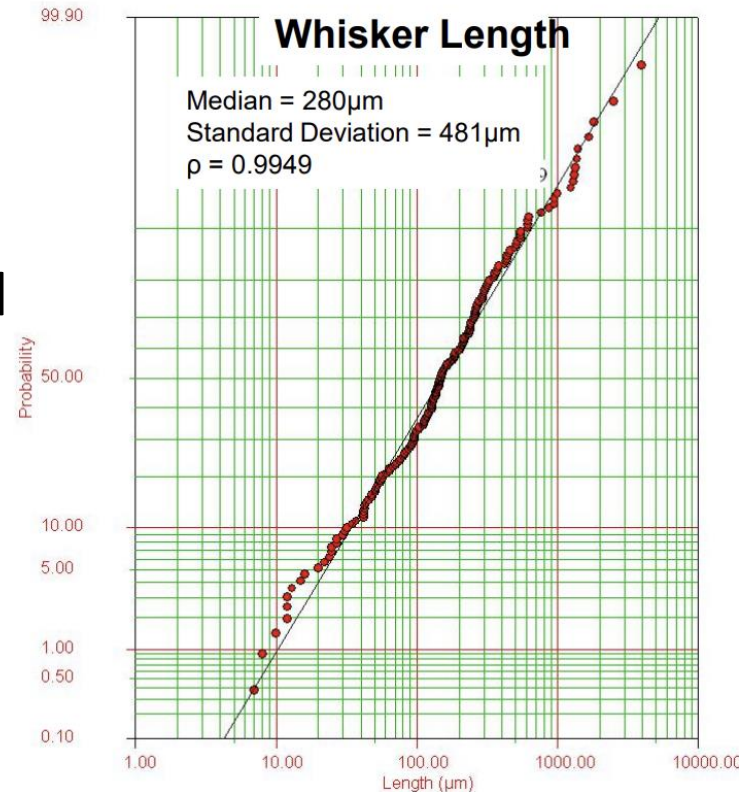


Control Functionality Chart

Background Information

Whisker Data

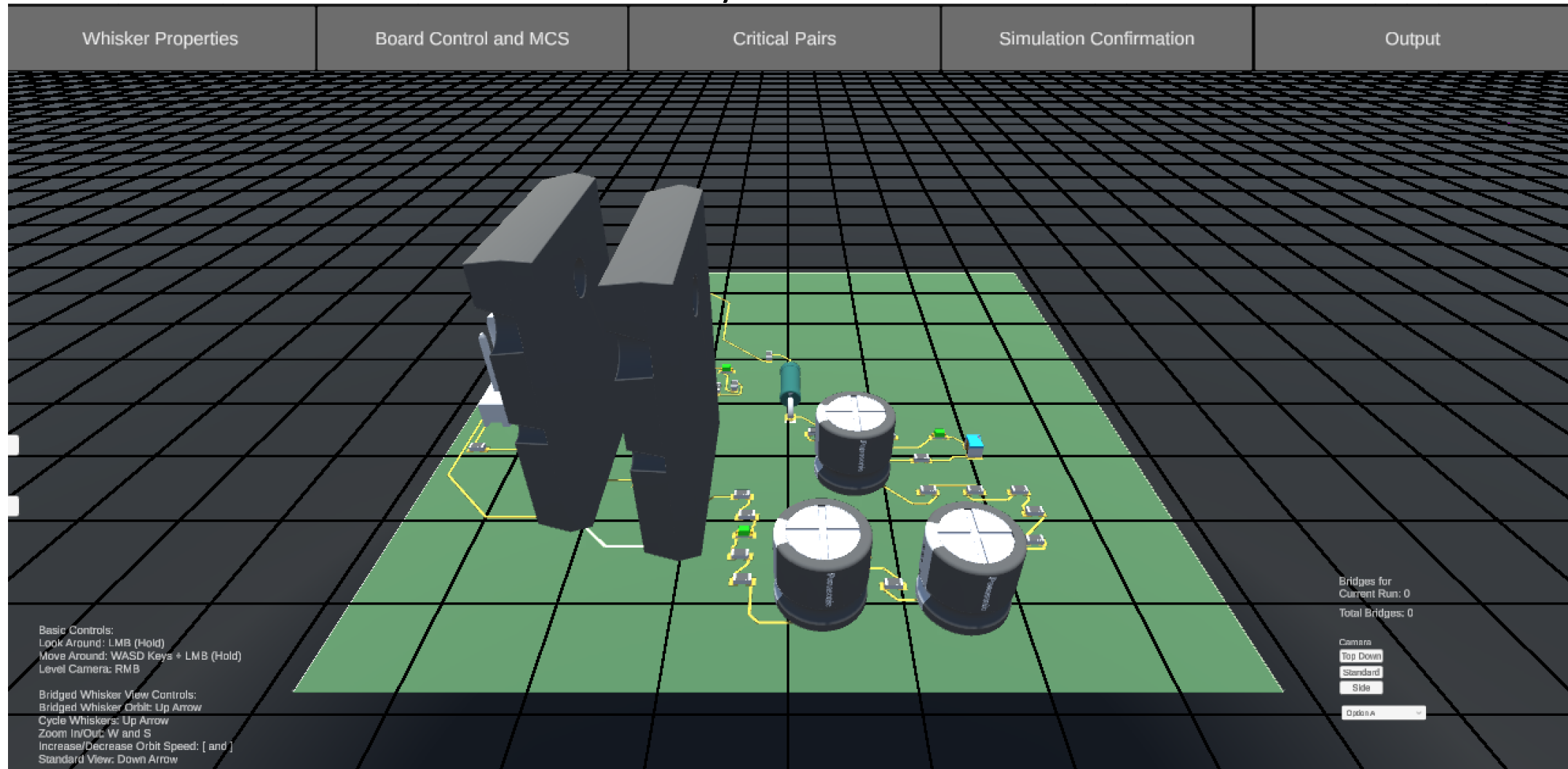
Example data for tin whisker length and thickness distributions (collected for 187 whiskers on tin-plated brass that grew over 11 years of ambient storage) is shown in the lognormal graphs to the right.



Example of Tin Whisker Length and Thickness Distributions [1]

Background Information

Unity Simulation



Background Information

User Interface: Whisker Properties

[Average Dimensions]

Length Mu (μm)

Width Mu (μm)

Whisker Material

Distribution Select

[Standard Deviation]

Length Sigma (μm)

Width Sigma (μm)

of Whiskers

Conductive Material Selection

The screenshot shows a web-based user interface titled "Whisker Properties | Conductive Materials". It features several input fields for configuring whisker parameters. The fields are arranged in a grid-like fashion. The first row contains "Length Mu [μm]" with a value of 4000, "Length Sigma [μm]" with a value of 1.18, and "# Of Whiskers" with a value of 100. The second row contains "Width Mu [μm]" with a value of 100, "Width Sigma [μm]" with a value of 0.67, and "Conductive Material Selection" with a placeholder text "Enter text...". The third row contains "Whisker Material" with a dropdown menu showing "Tin" and "Distribution Select" with a dropdown menu showing "Normal". The interface has a dark background with a subtle grid pattern. At the bottom, there are two large buttons labeled "Whisker Properties" and "Board C".

Length Mu [μm]	Length Sigma [μm]	# Of Whiskers
4000	1.18	100
Width Mu [μm]	Width Sigma [μm]	Conductive Material Selection
100	0.67	Enter text...
Whisker Material		
Tin		
Distribution Select		
Normal		

Whisker Properties Board C

Background Information

User Interface: Board Control & Monte Carlo Simulation

Select Gravity (Earth, Moon, Mars, Custom)

of Iterations

Enter Directory and
File Name

Save Inputs

Load Inputs

Add Walls (mm)

Wall Height

Toggleable Heatmap

The screenshot shows a web-based user interface for a Monte Carlo simulation. The background is dark with a grid pattern. At the top left, there is a 'Select Gravity' dropdown menu with 'Earth' selected. To its right is the title 'Monte Carlo | Data'. Below the title, there is a '# of Iterations' input field with the value '4'. To the right of this is a 'Save Inputs' button. Below the iterations field is a text input for 'Enter Directory and File Name:' with the example 'Ex: C:\JayBrusse'. To the right of this is a 'Load Inputs' button. Below the directory field is another text input with the example 'Ex: whisker_data', followed by a 'Save' button. To the right of the 'Save' button is a 'Heatmap' checkbox, which is currently unchecked. To the right of the 'Load Inputs' button is an 'Add Walls (mm)' button. Below this is a 'Wall Height' input field with the value '100' and a 'Colling' checkbox, which is checked. At the bottom of the interface, there are two large buttons: 'Whisker Properties' on the left and 'Board Control and MCS' on the right.

Background Information

User Interface: Board Control & Monte Carlo Simulation

Rotate Board (X, Y, Z) [degree]

Spin Board (X, Y, Z) [degree/sec]

Shock Amplitude [N]

Vibration Amplitude [N], Frequency [Hz], Duration [s]

The screenshot displays a user interface for board control and simulation, set against a dark background with a grid pattern. The interface is organized into several sections:

- Rotation Controls:** Three rows of controls for rotating the board. Each row includes a label (e.g., "Rotate Board X [degree]"), a text input field (e.g., "X"), and a dropdown menu (e.g., "X").
- Spinning Controls:** Three rows of controls for spinning the board. Each row includes a label (e.g., "Spin Board X [deg/sec]"), a text input field (e.g., "Enter text..."), and a dropdown menu (e.g., "X").
- Shock Controls:** A section with a "Shock" label, a checkbox labeled "Use in", a text input field for "Amplitude [N]" (containing "Enter text..."), and a dropdown menu (e.g., "X").
- Vibration Controls:** A section with a "Vibration" label, a checkbox labeled "Use in Simulation", and three text input fields for "Amplitude [N]" (containing "Enter text..."), "Frequency [Hz]" (containing "Enter text..."), and "Duration [s]" (containing "Enter text...").
- Action Buttons:** Two large buttons labeled "Shock" and "Vibrate" are positioned on the right side of the interface.
- Reset Board:** A checkbox labeled "Reset Board" is located below the spinning controls.

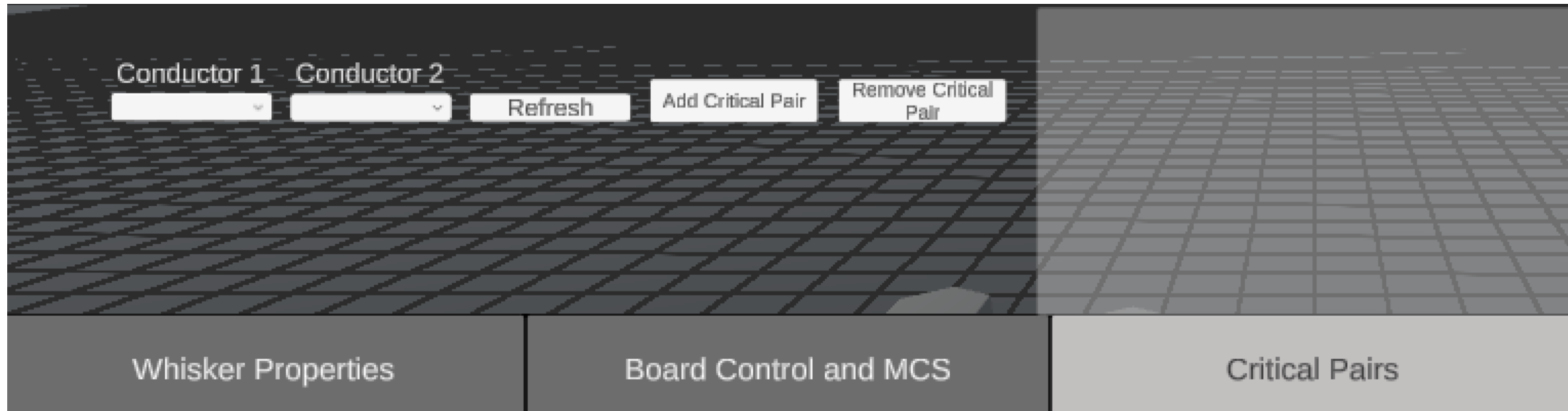
Background Information

User Interface: Critical Pairs

Conductors 1 & 2

Refresh

Add/Remove Critical Pair



Background Information

User Interface: Simulation Confirmation

Spawn Whiskers
Reload
Reset Gravity

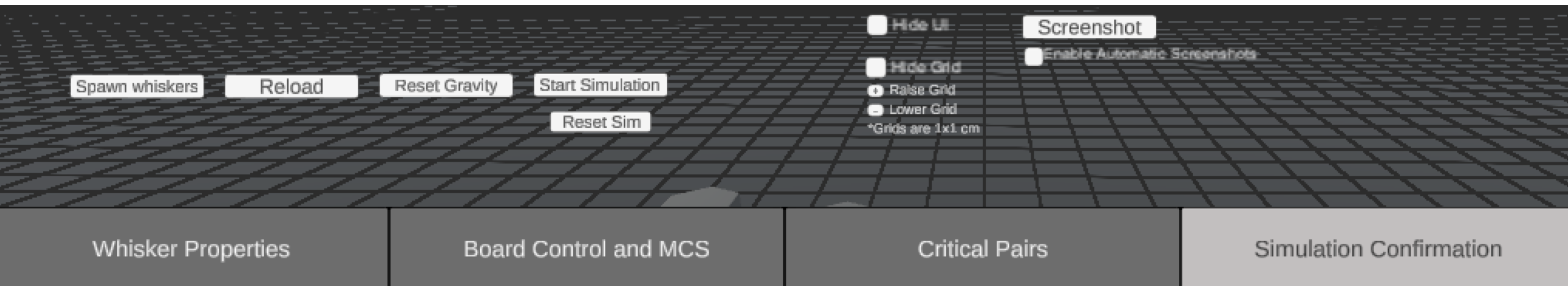
Start Simulation
Reset Simulation

Hide UI

Hide, Raise, Lower Grid
*Grids are 1x1 cm

Screenshot

Enable Automatic Screenshots

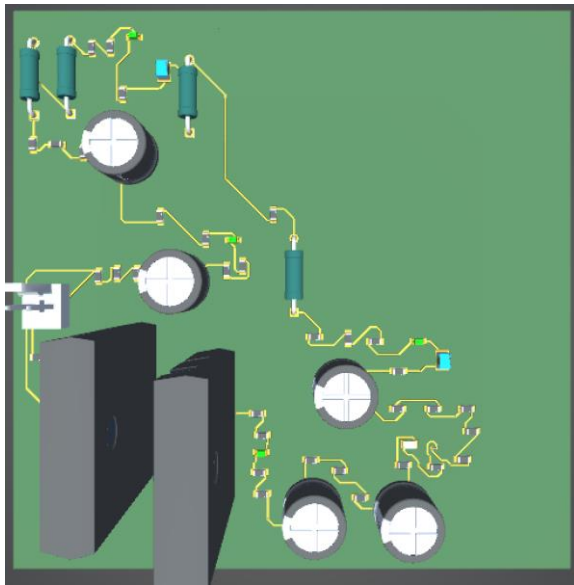


Background Information

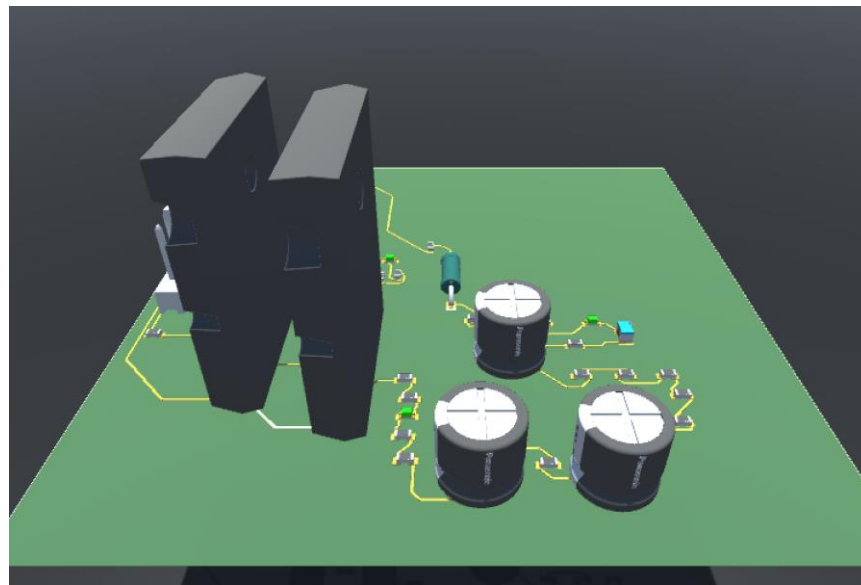
Camera

The camera for the simulation can be controlled by the user with their mouse and keyboard, but there are three pre-set camera angles the user can choose from:

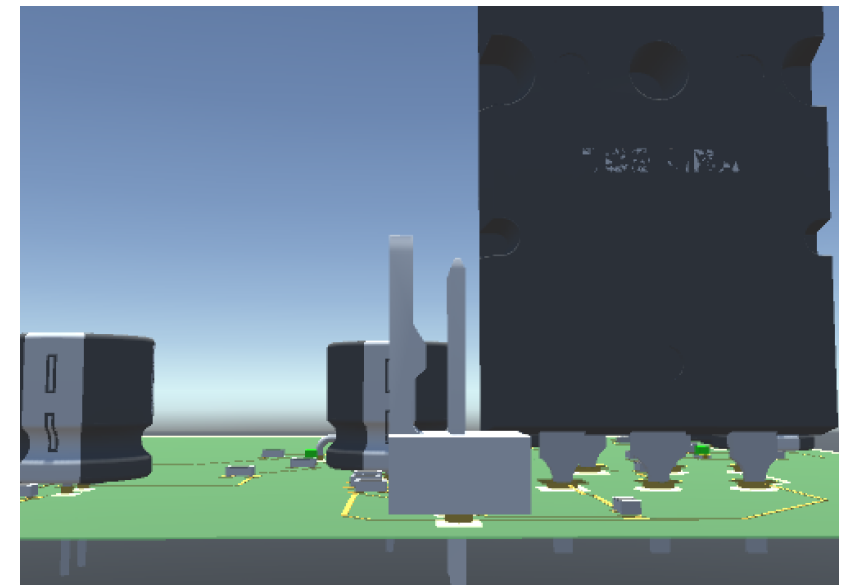
Top-Down



Standard (Default)

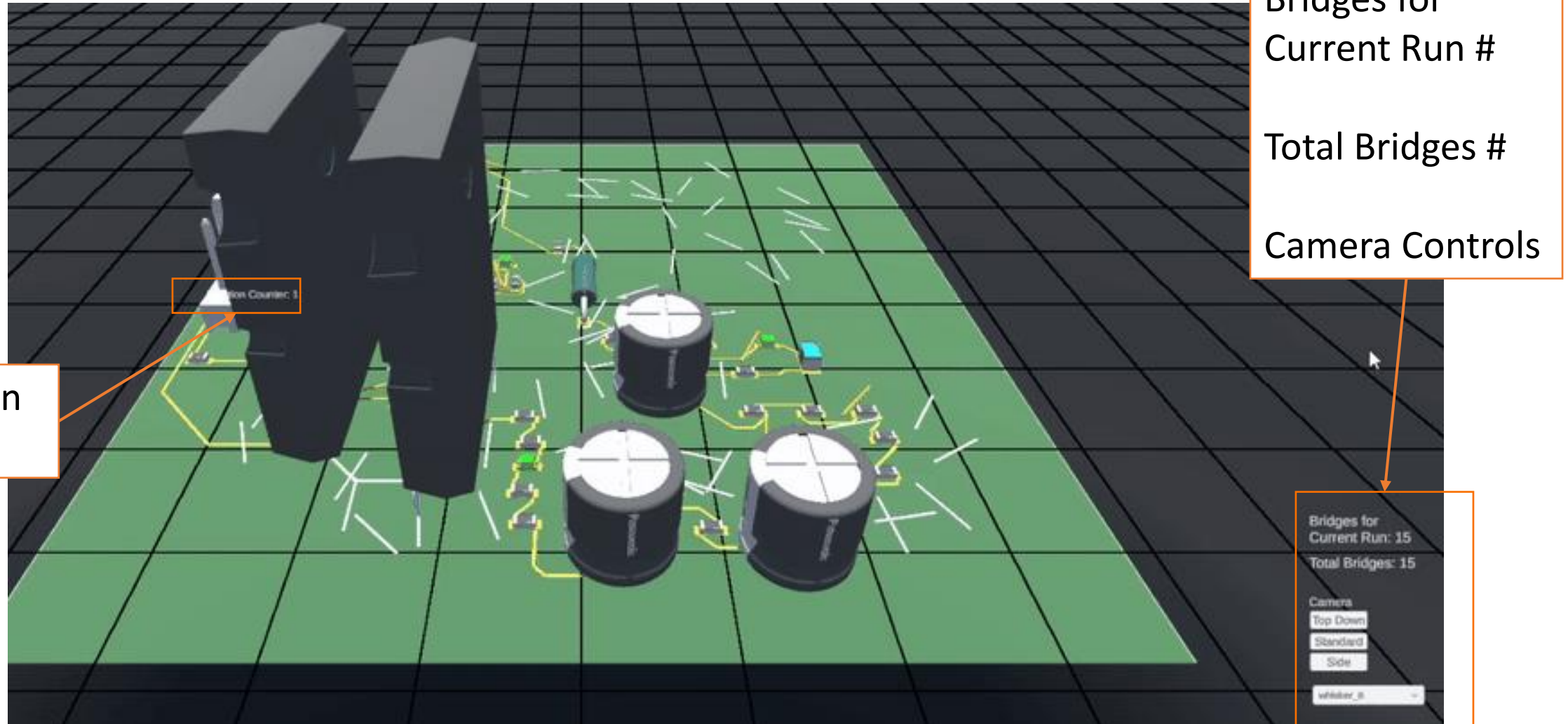


Side



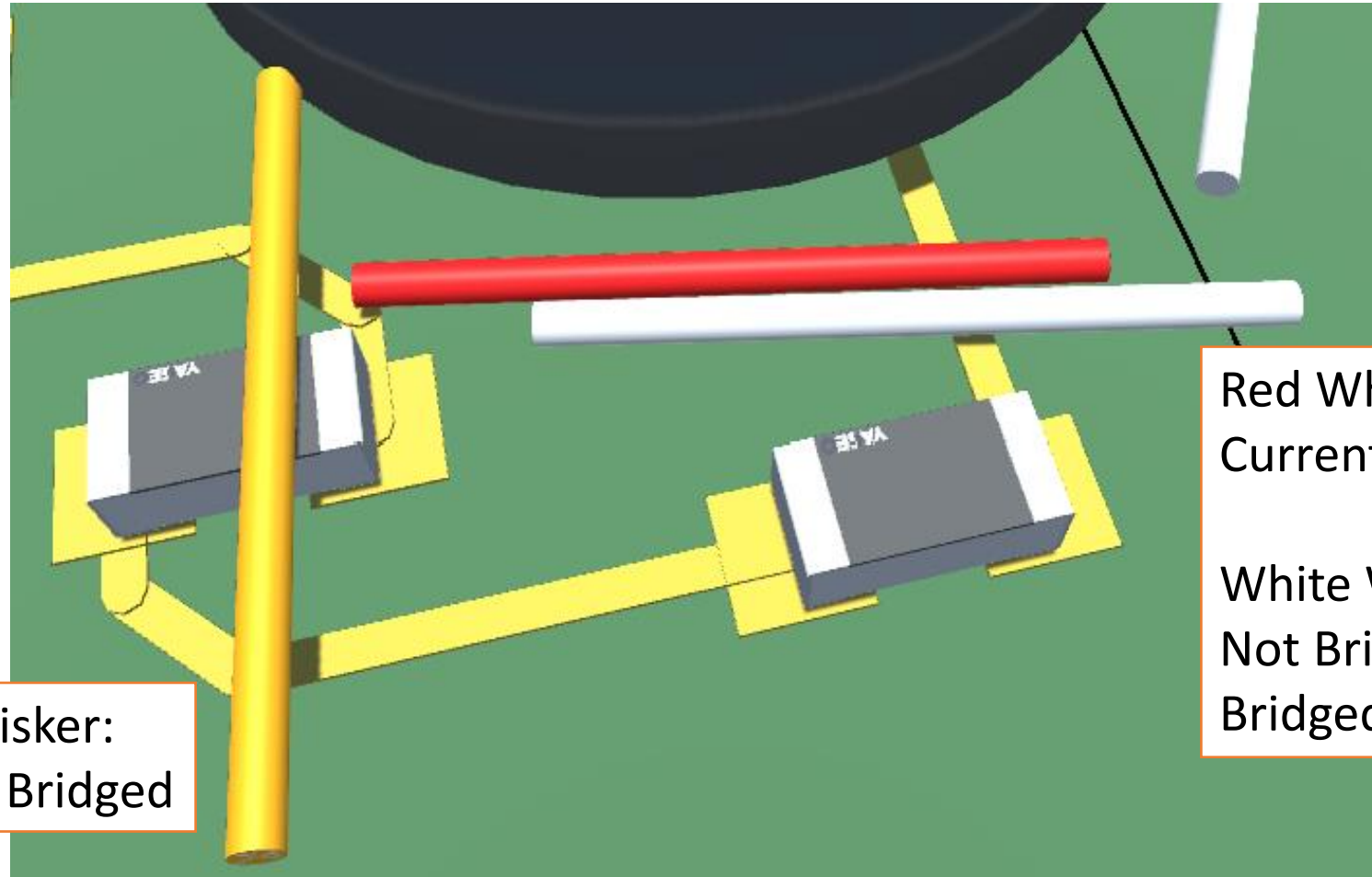
Background Information

Simulation (GIF of Two Iterations)



Background Information

Whiskers During Simulation



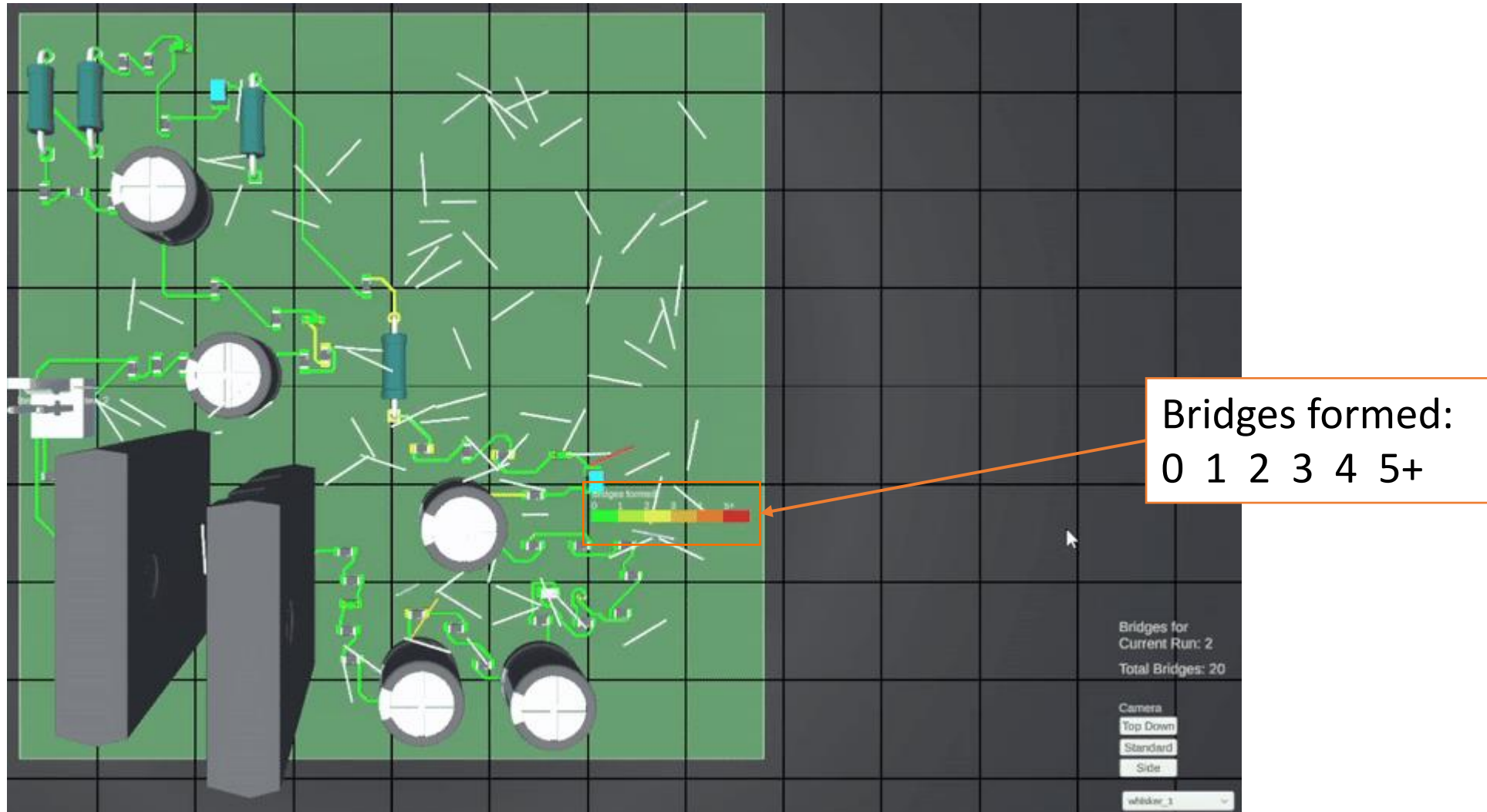
Yellow Whisker:
Previously Bridged

Red Whisker:
Currently Bridging

White Whisker:
Not Bridging & Has Not
Bridged Before

Background Information

Simulation Heatmap (GIF of Two Iterations)



Background Information

Simulation Output (Unity)

Whisker #	Length (um)	Diameter (um)	Resistance (ohm)	Iteration	Conductor 1	Conductor 2
Board Control and MCS			Critical Pairs		Simulation Confirmation	
25	3999.539	99.84045	0.0556844	1	R10	NetR9_2
87	3999.741	100.3905	0.05507865	1	Q1	R2
48	4000.531	100.0376	0.05547891	1	R29	L2
66	4000.625	99.34013	0.05626199	1	R6	R5
28	3997.878	100.0325	0.05544778	1	R6	NetC1_2
94	3999.284	99.39487	0.05618118	1	NetR16_2	R17
36	3998.853	99.28466	0.05629991	1	NetL1_1	NetD2_1
76	4001.261	100.3666	0.05512579	1	R15	NetR12_1
54	3999.222	99.8329	0.0556884	1	R32	NetR31_2
57	3998.696	100.4456	0.05500381	1	R7	NetC1_1
60	4000.404	99.83982	0.05569714	1	R19	NetD2_1
61	4003.138	100.4083	0.05510587	1	R7	NetR7_2
37	3998.822	100.7604	0.05466241	2	C2	NetR11_2
68	3998.306	101.9977	0.0533374	2	Q1	NetQ1_1
12	3999.702	100.0685	0.05543313	2	NetC1_1	NetR7_2
78	3998.138	99.39001	0.05617058	2	NetC1_2	R6
53	4000.4	100.4902	0.05497841	2	NetC4_2	R17
100	3999.466	99.44799	0.05612373	2	R29	NetR29_2
17	3998.444	100.3501	0.0551051	2	GND	NetQ1_1
60	3997.822	101.0631	0.0543219	2	R37	NetC6_1
9	4002.269	99.68561	0.05589563	2	R22	NetR22_2
20	4000.569	99.7717	0.05577552	2	R28	NetD3_1
28	4000.063	100.627	0.05482448	2	R35	NetR35_2
95	3998.811	100.7568	0.05466614	2	R34	NetR33_2
27	4000.442	99.44227	0.0556122	2	R16	NetR16_2

Output

Load CSV

Close

Background Information

Simulation Output (Excel)

All Whiskers						
Whisker #	Length (um)	Width (um)	Resistance (ohm)	Iteration		
1	3999.173	99.80888	0.05571452	1		
2	4001.038	99.66998	0.05589598	1		
3	3999.219	100.0819	0.05541163	1		
4	3998.486	99.58105	0.05596013	1		
5	4000.987	99.94893	0.05558369	1		
6	4001.222	99.6096	0.05596633	1		
7	4001.122	100.397	0.05509054	1		
8	3997.827	99.39907	0.05615598	1		
9	4000.042	98.88059	0.05677787	1		
10	4000.345	100.0352	0.05547893	1		
11	3999.434	99.8668	0.05565355	1		

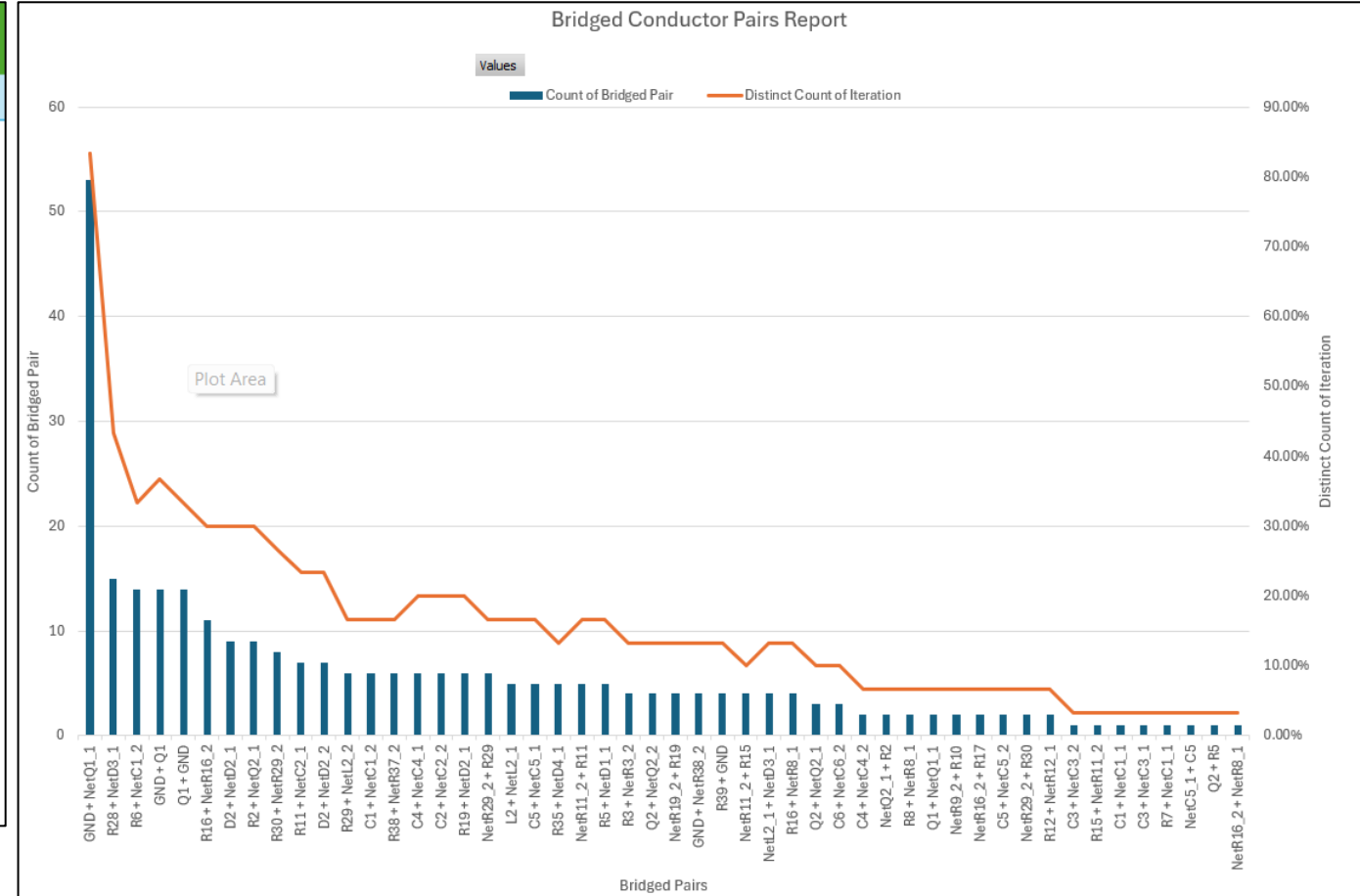
Bridged Whiskers									
Whisker #	Length (um)	Width (um)	Resistance (ohm)	Iteration	Conductor 1	Conductor 2	Bridged Pair		
318	3272.759	51.13706	0.1736917	1	GND	Q1	GND + Q1		
352	4102.102	55.02091	0.1880561	1	GND	Q1	GND + Q1		
26	4421.396	53.90409	0.2111799	1	Q1	GND	Q1 + GND		
43	3747.569	56.62005	0.1622355	1	R28	NetD3_1	R28 + NetD3_1		
56	2372.752	45.65566	0.1579791	1	C4	NetC4_1	C4 + NetC4_1		
198	6977.418	59.15441	0.2767306	1	Q2	NetQ2_2	Q2 + NetQ2_2		
117	6817.568	51.01569	0.3635453	1	GND	NetQ1_1	GND + NetQ1_1		
9	6449.581	33.88324	0.7796476	1	NetR11_2	C2	NetR11_2 + C2		
381	2518.593	63.03785	0.08796133	1	P30	NetP30_2	P30 + NetP30_2		

Background Information

Simulation Output (Excel)

Bridged Pair Analysis

Bridged Pairs	Count of Bridged Pair	Distinct Count of Iteration
GND + NetQ1_1	53	83.33%
R28 + NetD3_1	15	43.33%
R6 + NetC1_2	14	33.33%
GND + Q1	14	36.67%
Q1 + GND	14	33.33%
R16 + NetR16_2	11	30.00%
D2 + NetD2_1	9	30.00%
R2 + NetQ2_1	9	30.00%
R30 + NetR29_2	8	26.67%
R11 + NetC2_1	7	23.33%
D2 + NetD2_2	7	23.33%
R29 + NetL2_2	6	16.67%
C1 + NetC1_2	6	16.67%
R38 + NetR37_2	6	16.67%
C4 + NetC4_1	6	20.00%
C2 + NetC2_2	6	20.00%
R19 + NetD2_1	6	20.00%
NetR29_2 + R29	6	16.67%
L2 + NetL2_1	5	16.67%
C5 + NetC5_1	5	16.67%



Design Elements

Overview

- Add in a way to simulate Electrostatic Attraction that will pull whiskers at multiple points at varying strengths.
- Create Standardized Testing Profiles including Acceleration, Shock, and Vibration activation with a Demonstration Profile.
- Upgrade current Heatmap Output
- Improve PCB import process

Design Elements

Electrostatic Attraction

Electrostatic Attraction is the measure of which electric point charges exhibit forces pushing them in the direction of **each other**.

The example picture shows a cat, whose fur is **charged** with static electricity, is **attracting** foam peanuts with electrostatic attraction.



Design Elements

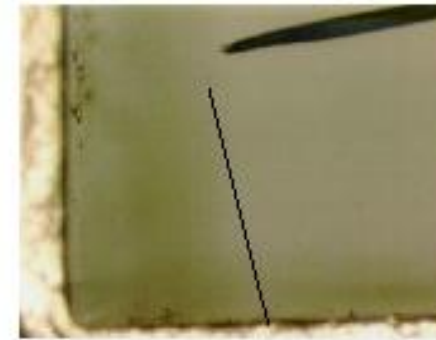
Electrostatic Attraction

- The magnitude of **electrostatic force** is measured by Coulomb's Law:

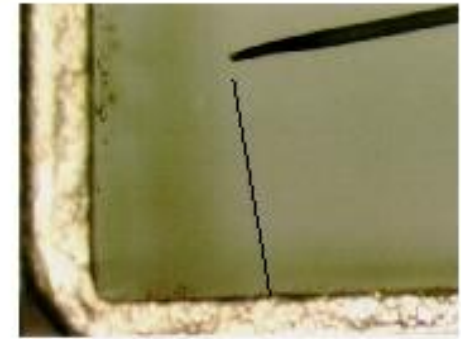
$$F = k \frac{q_1 q_2}{r^2}$$

Where $k = 8.99 * 10^9 \frac{N \cdot m^2}{C^2}$ (Coulomb's constant), q_1 and q_2 are the charges' magnitudes, and r is the distance between the charges.

- The circuit-boards in this project are **electrically active** and are generating **electric fields**, which Tin-Whiskers have shown to experience Electrostatic Attraction with.



Before the voltage applied



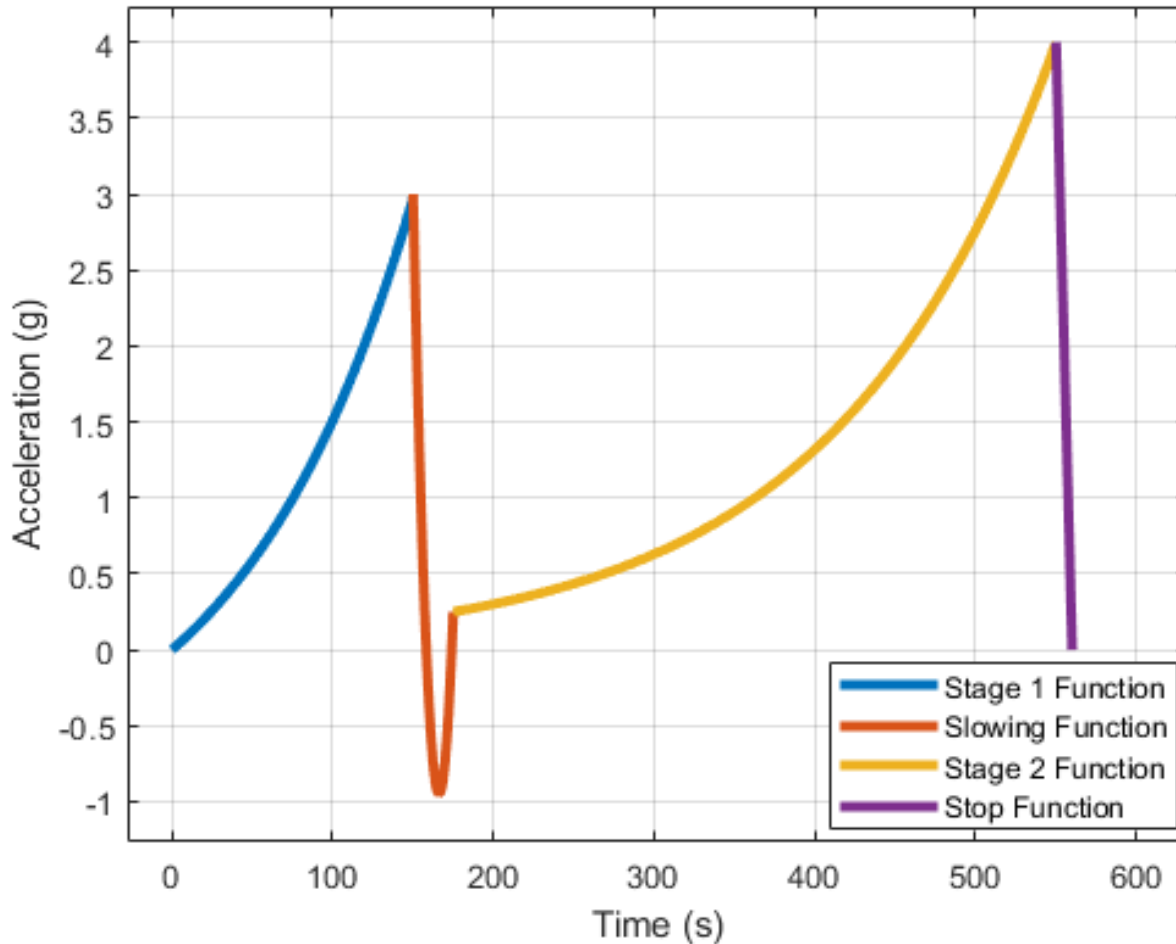
After the voltage applied

Example of a Tin-Whisker Bending in Reaction to an Electrical Field [5].

Design Elements

Simulation Profile

Acceleration Profile



Simulation Profile: Acceleration

- Acceleration magnitude will be input as a function of time.
- Acceleration direction will be an input of degrees as a function of time.
- Various functions can be used such as line, exponential, and polynomial.
- The Falcon 9 launch is a two-stage launch.

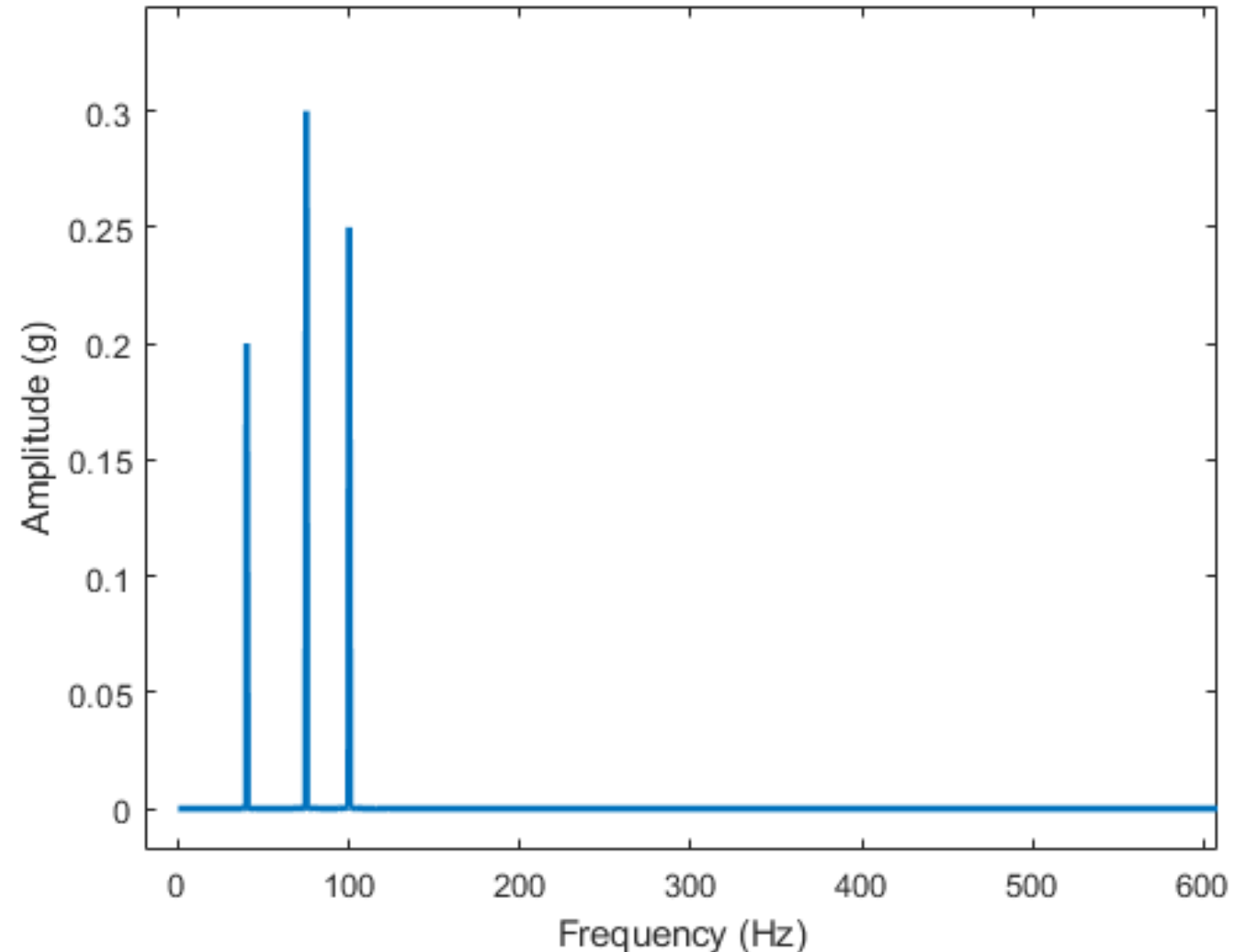
Design Elements

Simulation Profile: Vibration

- Sinusoidal vibration requires an amplitude and frequency input.
- Multiple vibrations can be input to be combined.
- The vibration will be applied over the three axis.
- Three vibrations are combined within the Falcon 9's specifications.

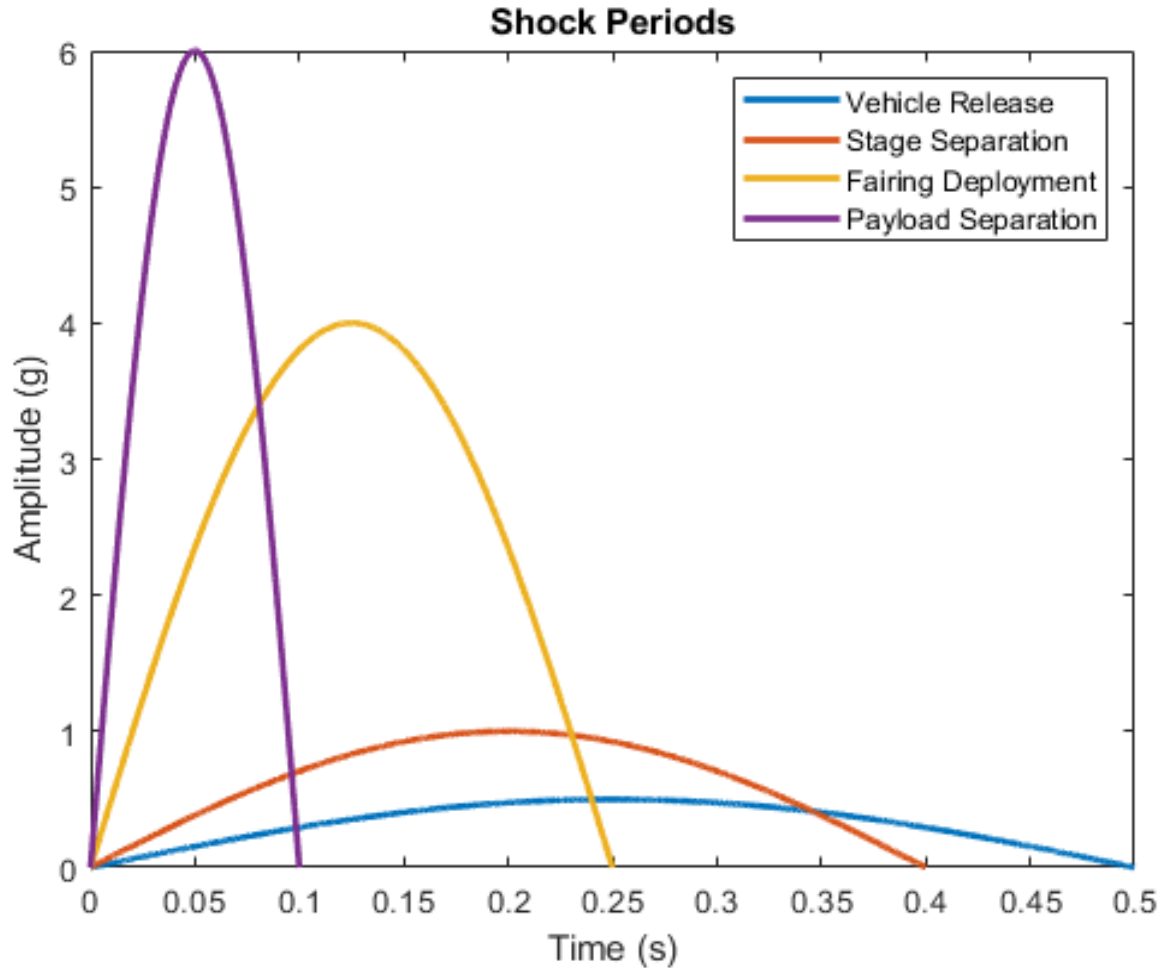
Simulation Profile

Vibration Analysis



Design Elements

Simulation Profile



Simulation Profile: Shock

- Shock input is the time period and amplitude.
- Shock will be added to vibration and affected in the same direction.
- Falcon 9 has four shock events: vehicle release, stage separation, fairing deployment, and payload separation.

Development, Testing, and Results

PCB Import

<https://docs.arduino.cc/hardware/leonardo/>

Home / Hardware / **UNO R4 WiFi**

UNO R4 WiFi

The Arduino UNO R4 WiFi is designed around the 32-bit microcontroller RA4M1 from [Renesas](#) while also featuring a ESP32 module for Wi-Fi® and Bluetooth® connectivity. Its distinctive 12x8 LED matrix makes it possible to prototype visuals directly on the board, and with a Qwiic connector, you can create projects plug-and-play style.

GET STARTED

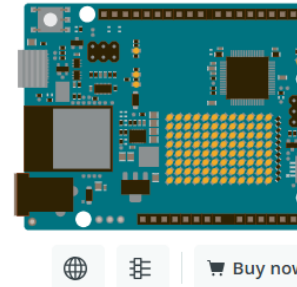
DOWNLOADABLE RESOURCES

 Pinout (PDF)

 Datasheet

 Schematics

 CAD Files



Features

Tutorials

Tech Specs

Compatibility

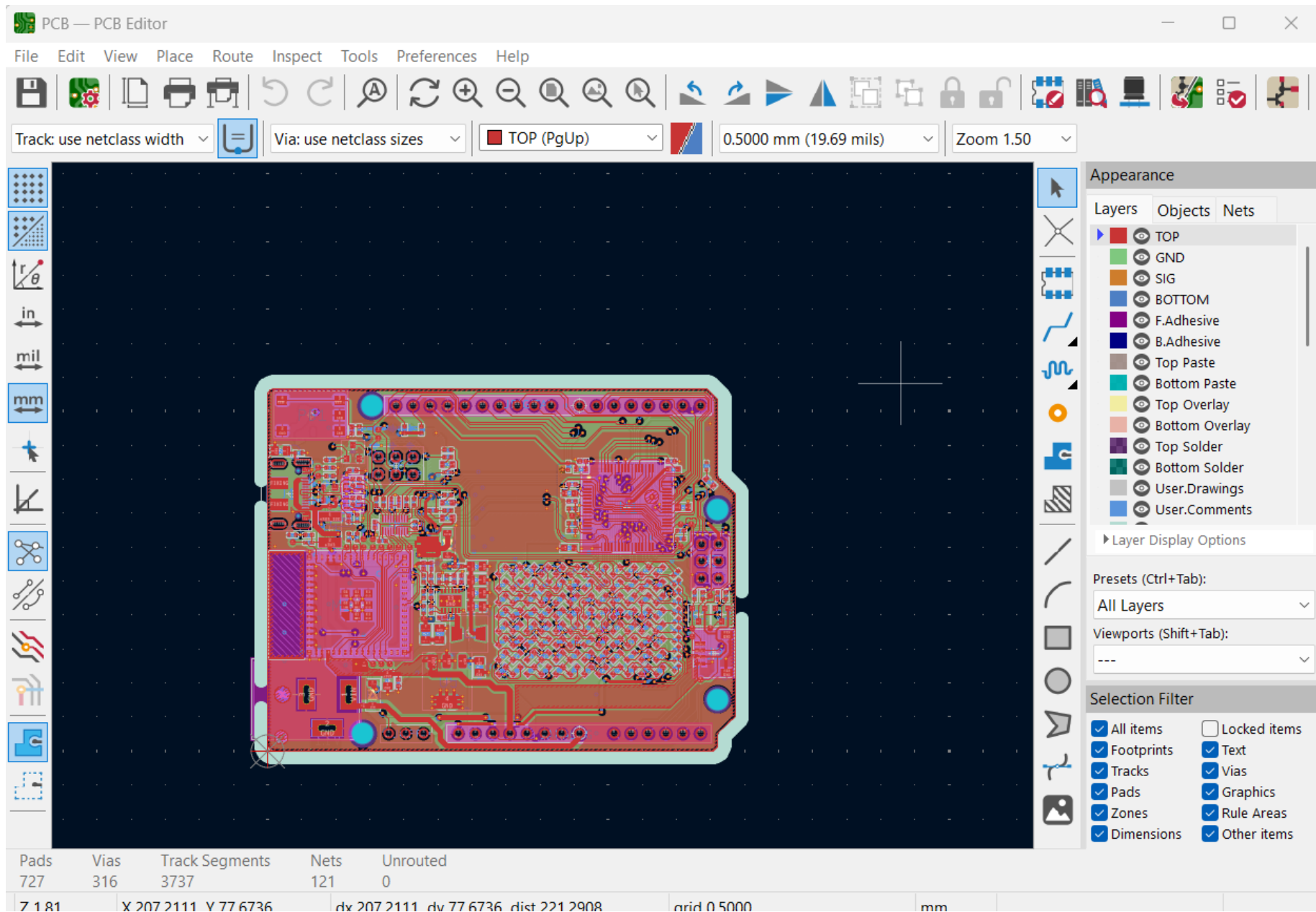
Suggested Libraries

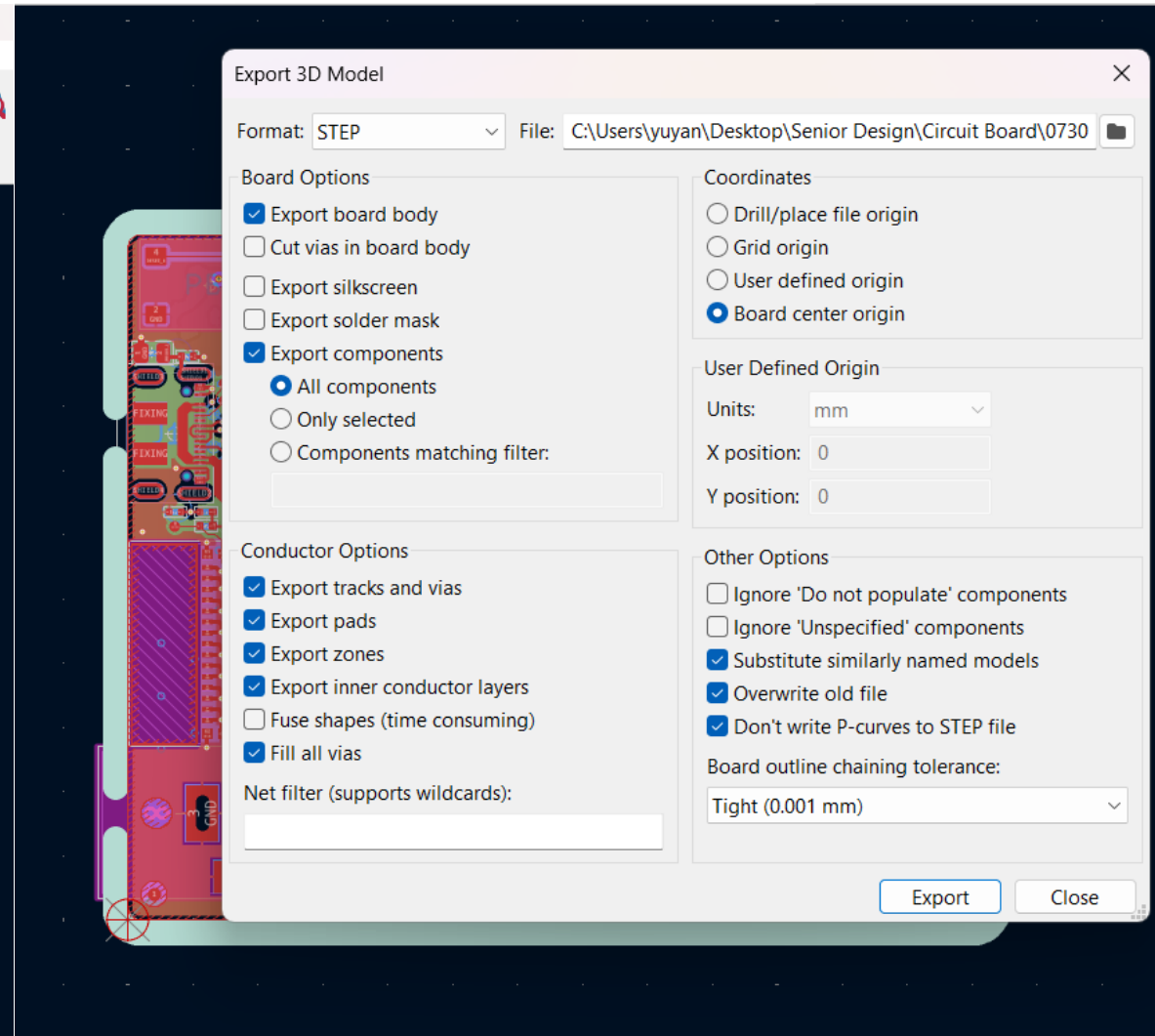
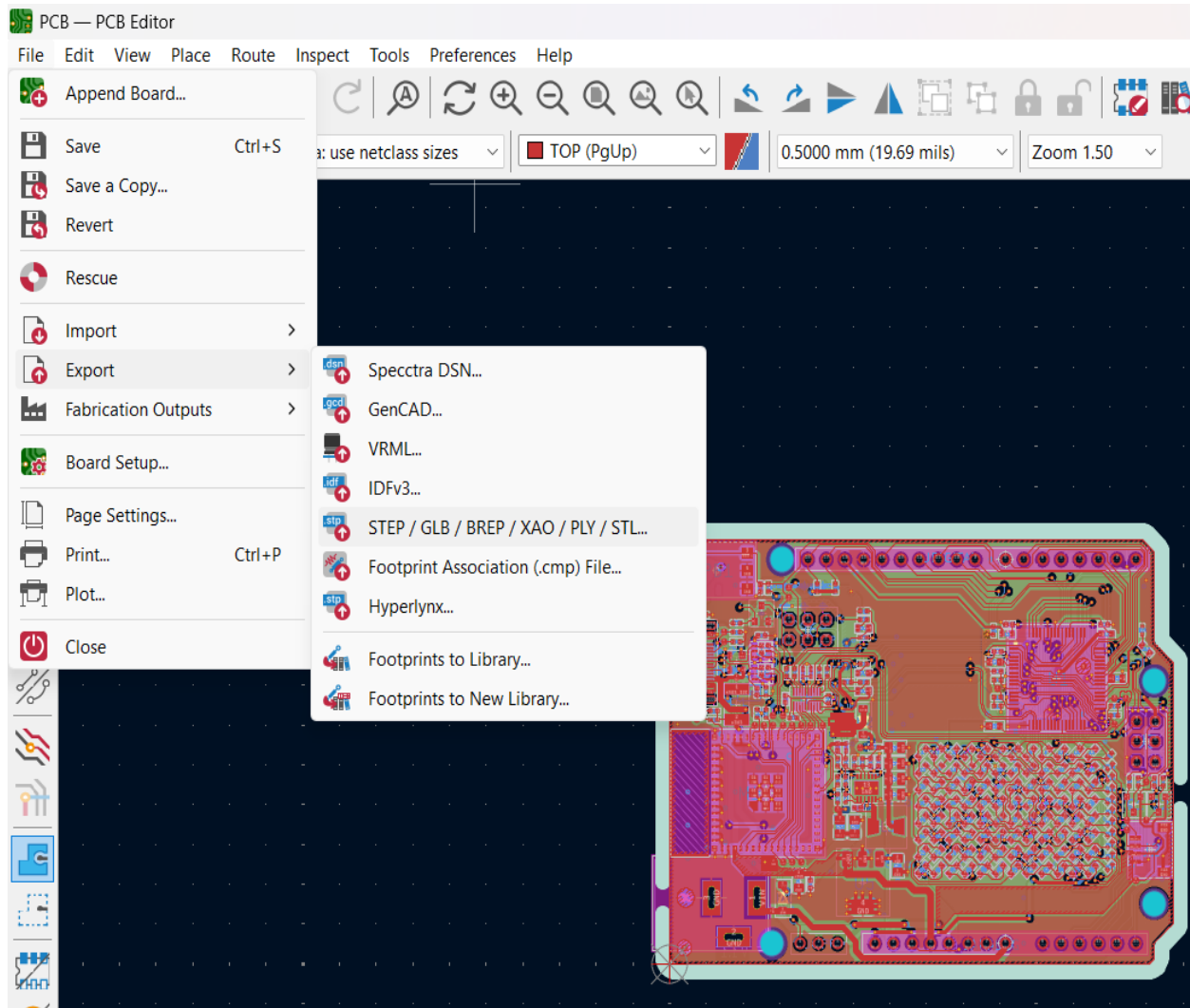
The MCU on the board is the high performance Renesas RA4M1 (Arm® Cortex®-M4) with a 48 MHz clock speed, 32 kB SRAM and 256 kB flash memory. This MCU features an RTC, a DAC and a CAN bus and has support for HID via USB.

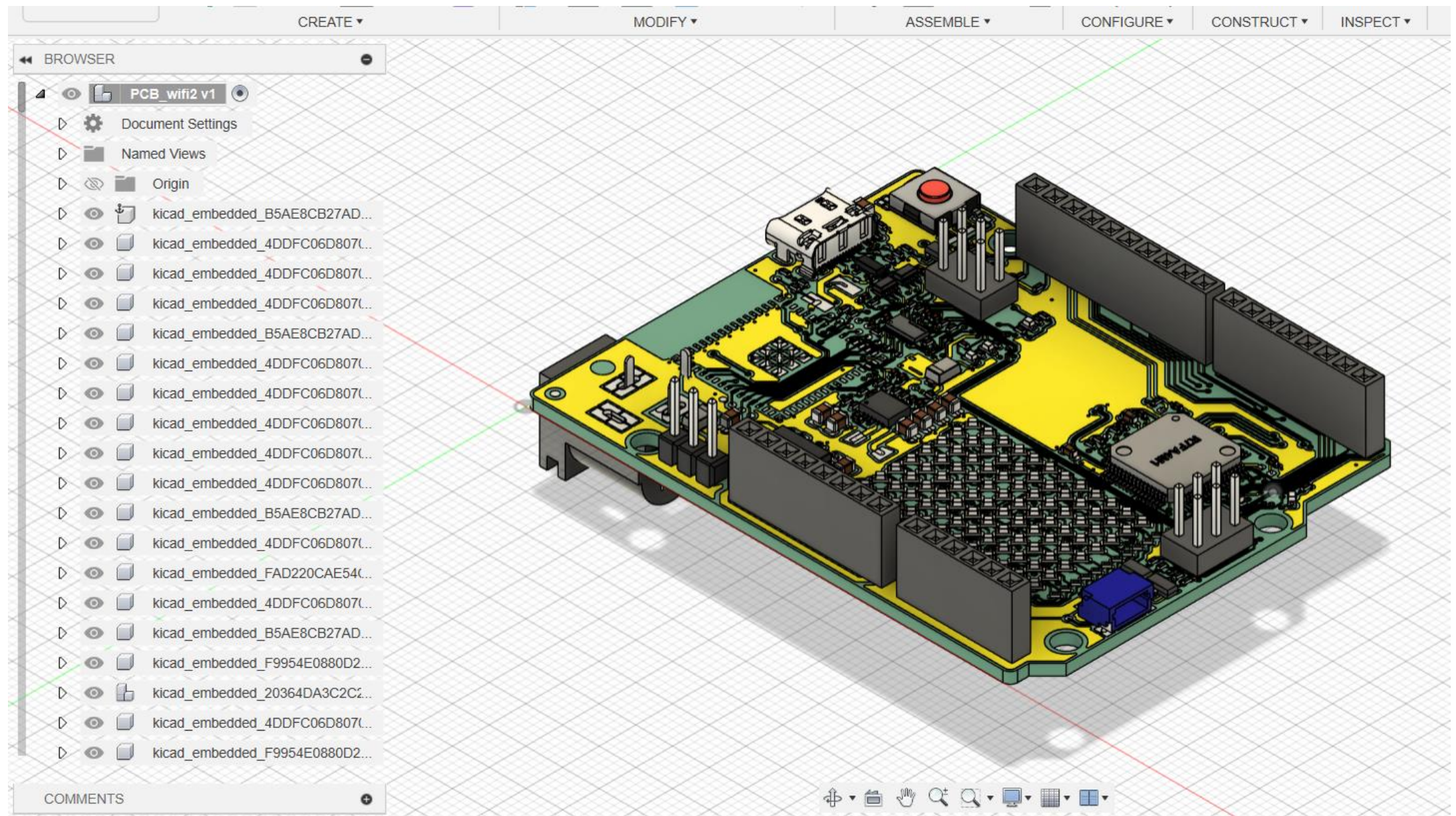
The UNO R4 WiFi also features an ESP32-S3 for Wi-Fi®/Bluetooth® connectivity, which can also be separately programmed via a specific header.



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Export



Name:

center v1

Type:

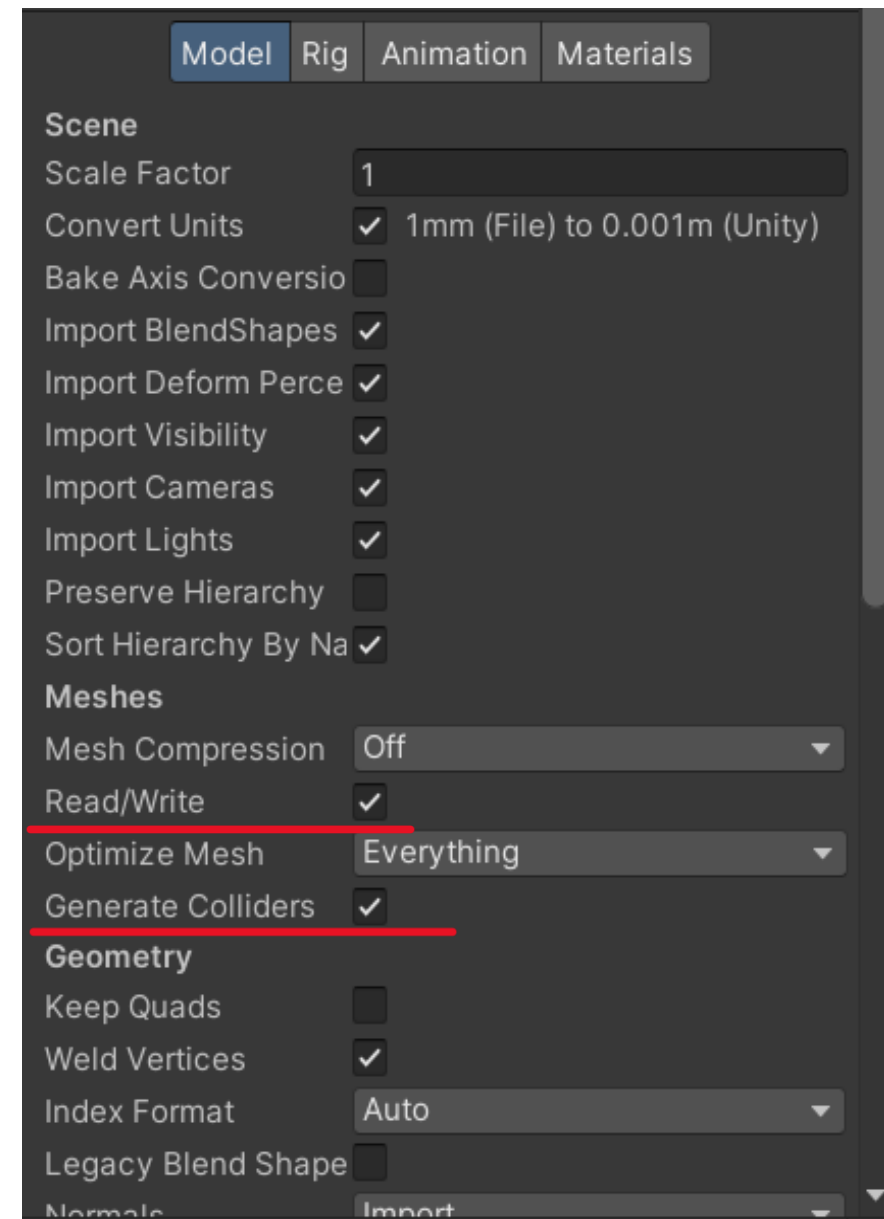
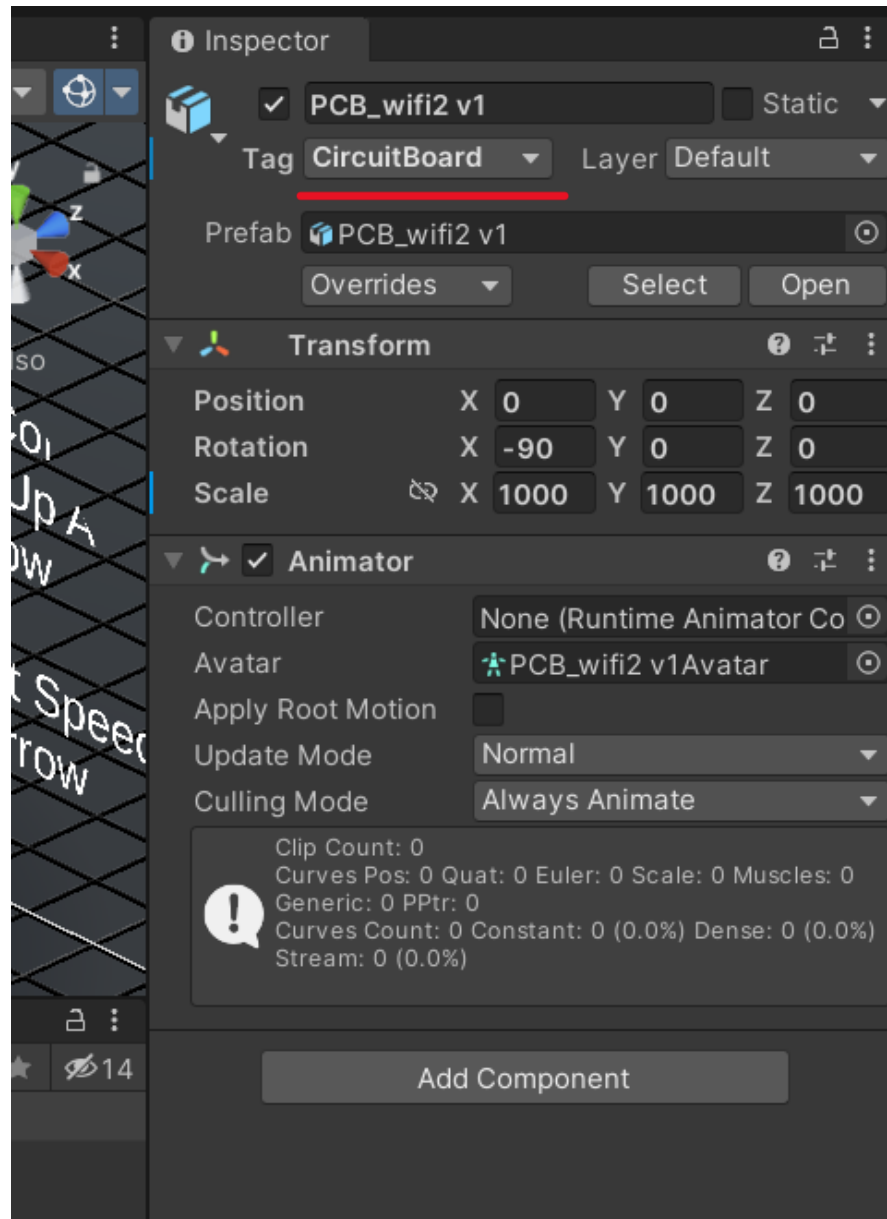
FBX Files (*.fbx) ▼

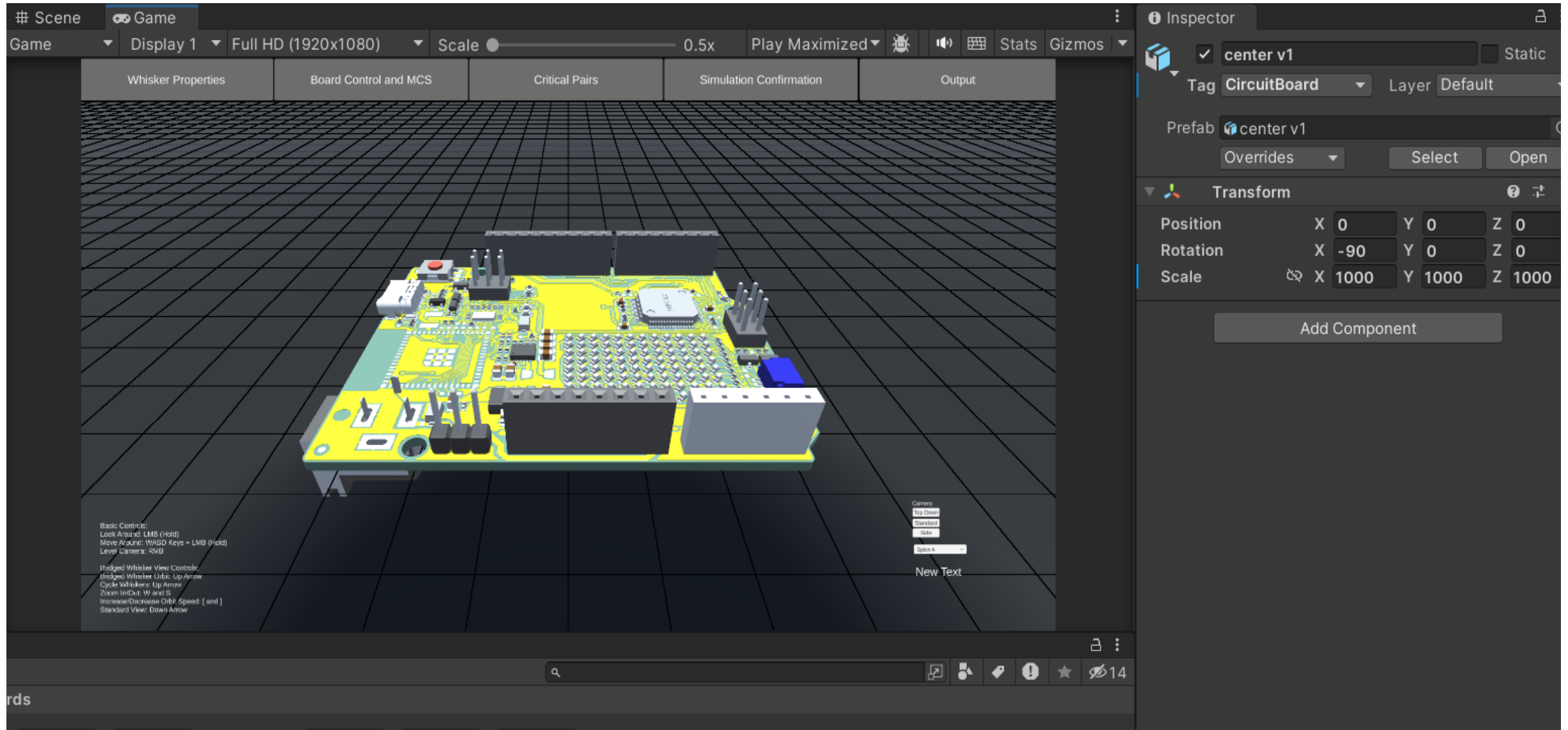
- Autodesk Fusion Archive Files (*.f3d)
- 3MF Files (*.3mf)
- Autodesk Inventor 2024 Files (*.ipt *.iam)
- DWG Files (*.dwg)
- DXF Files (*.dxf)
- FBX Files (*.fbx)**
- IGES Files (*.igs *.iges)
- OBJ Files (*.obj)
- SAT Files (*.sat)
- SketchUp Files (*.skp)

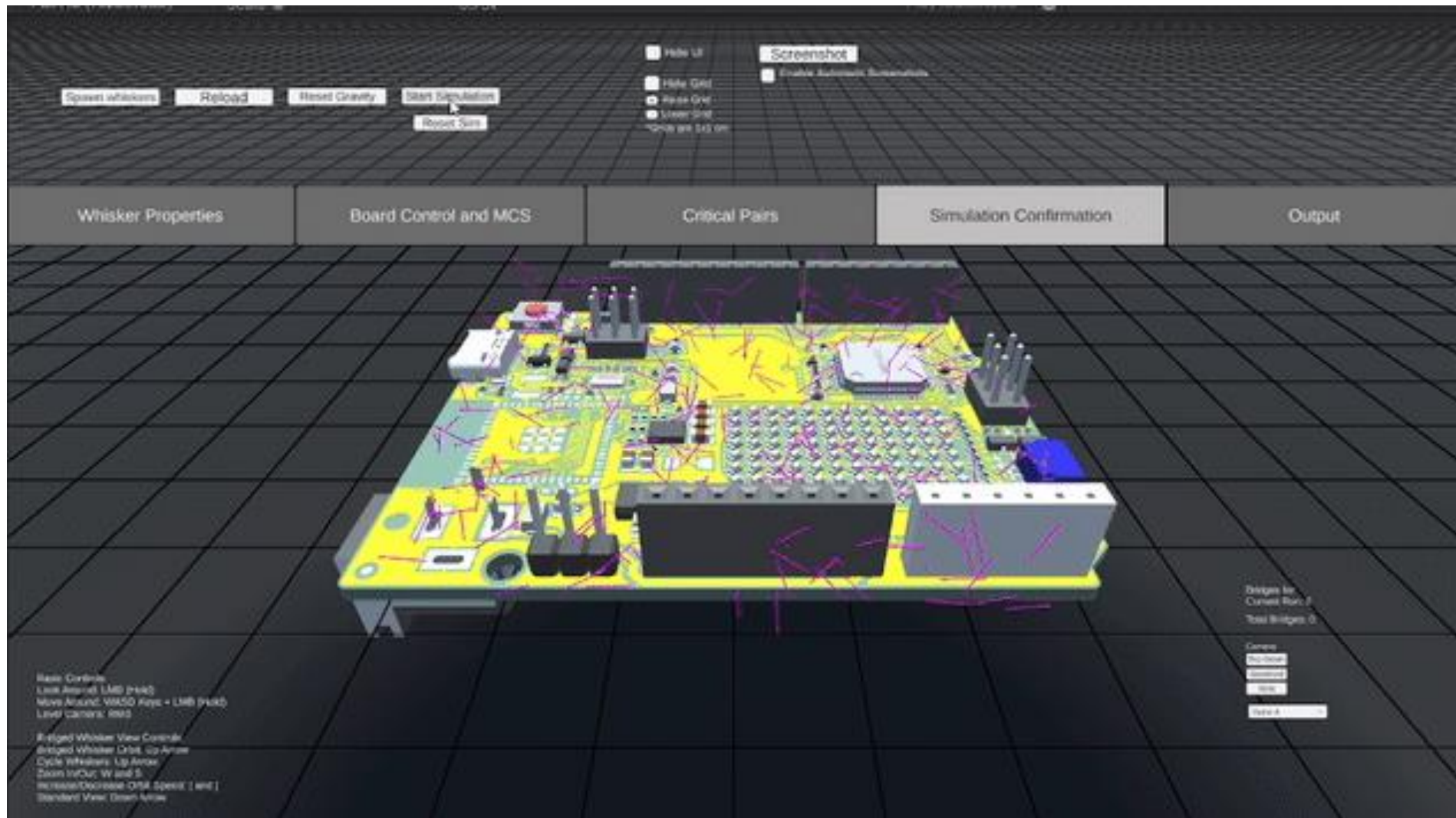
Cancel

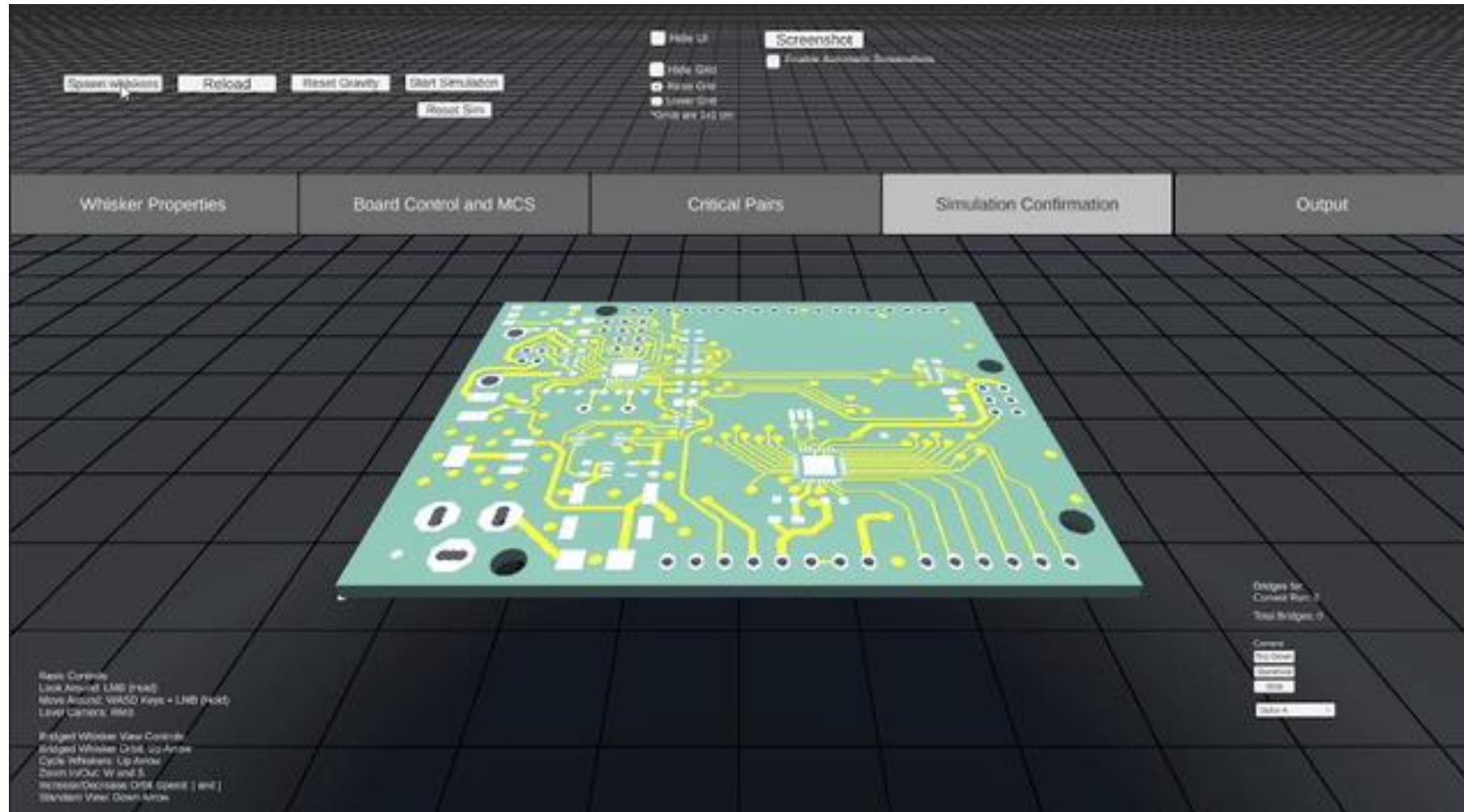
Export





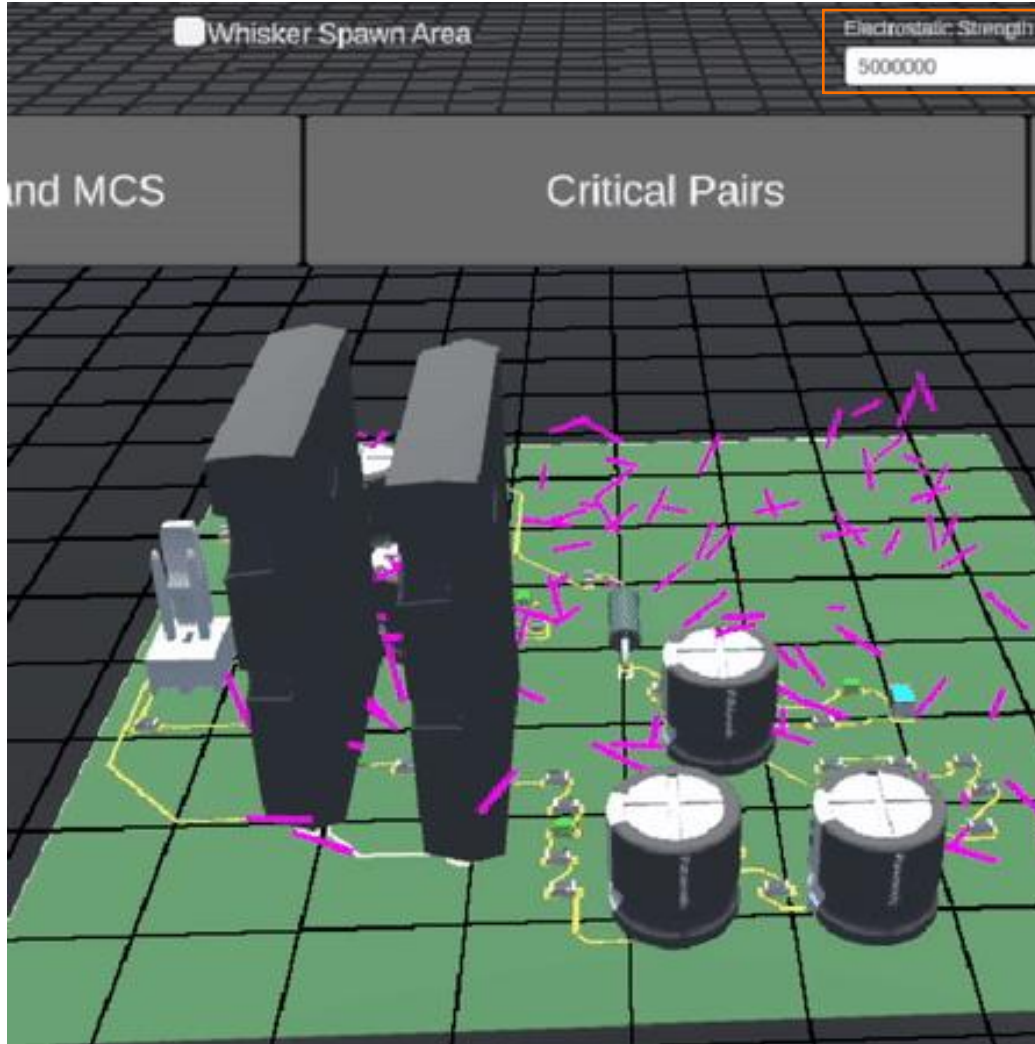






Development, Testing, and Results

Electrostatic Attraction



Phase One of
ES Attraction Development:
Dynamic Magnitude

$$F = k \frac{q_1 q_2}{r^2}$$

Electrostatic Strength $[Nm^2]$

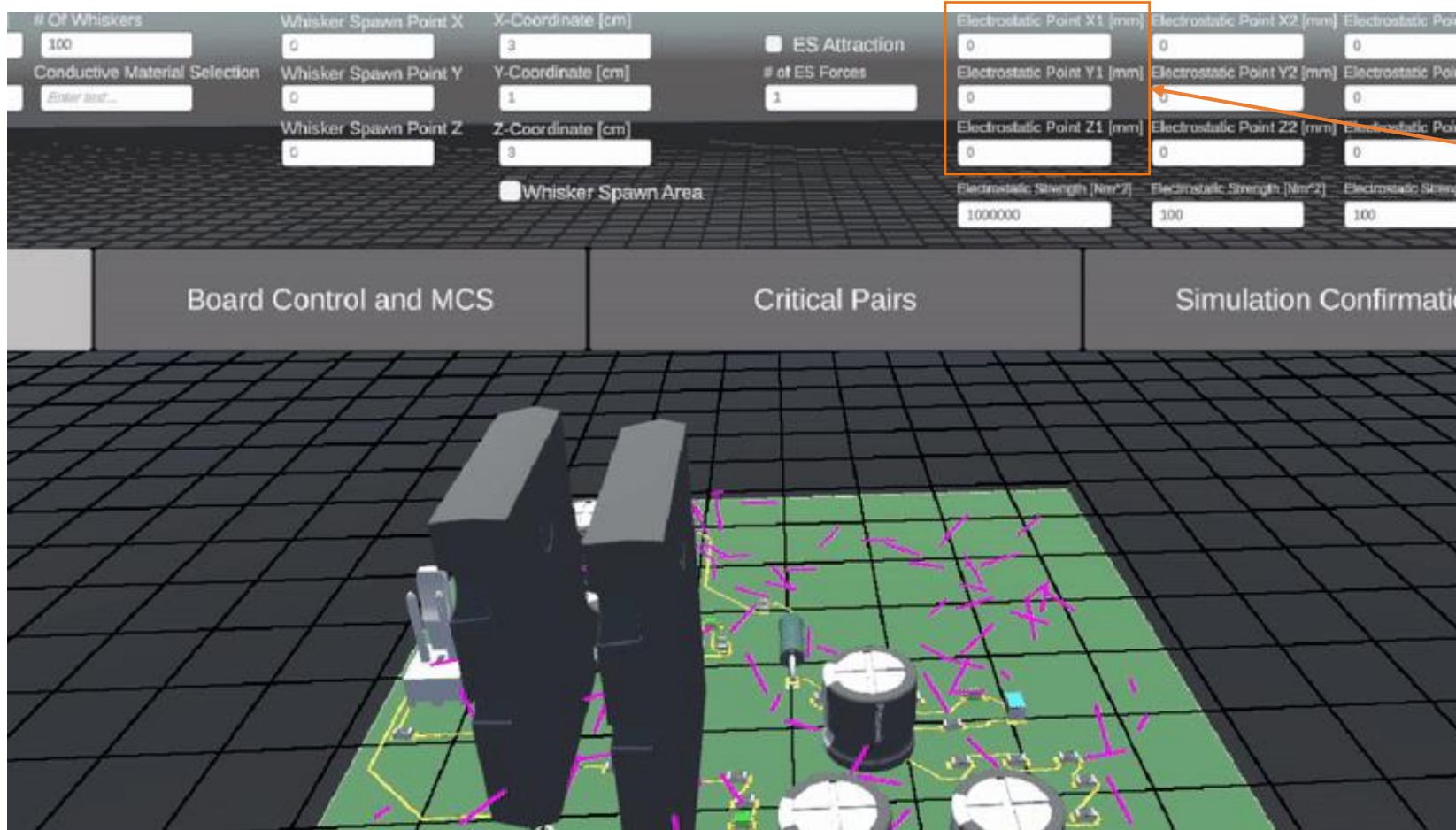
Electrostatic Strength = $k(q_1 q_2)$
Shown: Electrostatic Strength is changed
from positive (attracting) to negative
(repulsing) repeatedly

GIF

Development, Testing, and Results

Electrostatic Attraction

Phase Two of
ES Attraction Development:
Dynamic Position



Electrostatic Point X1 [mm]
Electrostatic Point Y1 [mm]
Electrostatic Point Z1 [mm]
Electrostatic Strength [Nm^2]

Shown: Starts at (0,0,0),
then ES Y1 becomes 25mm,
then ES X1 becomes 25mm,
then ES Z1 becomes 25mm

GIF

Development, Testing, and Results

Electrostatic Attraction

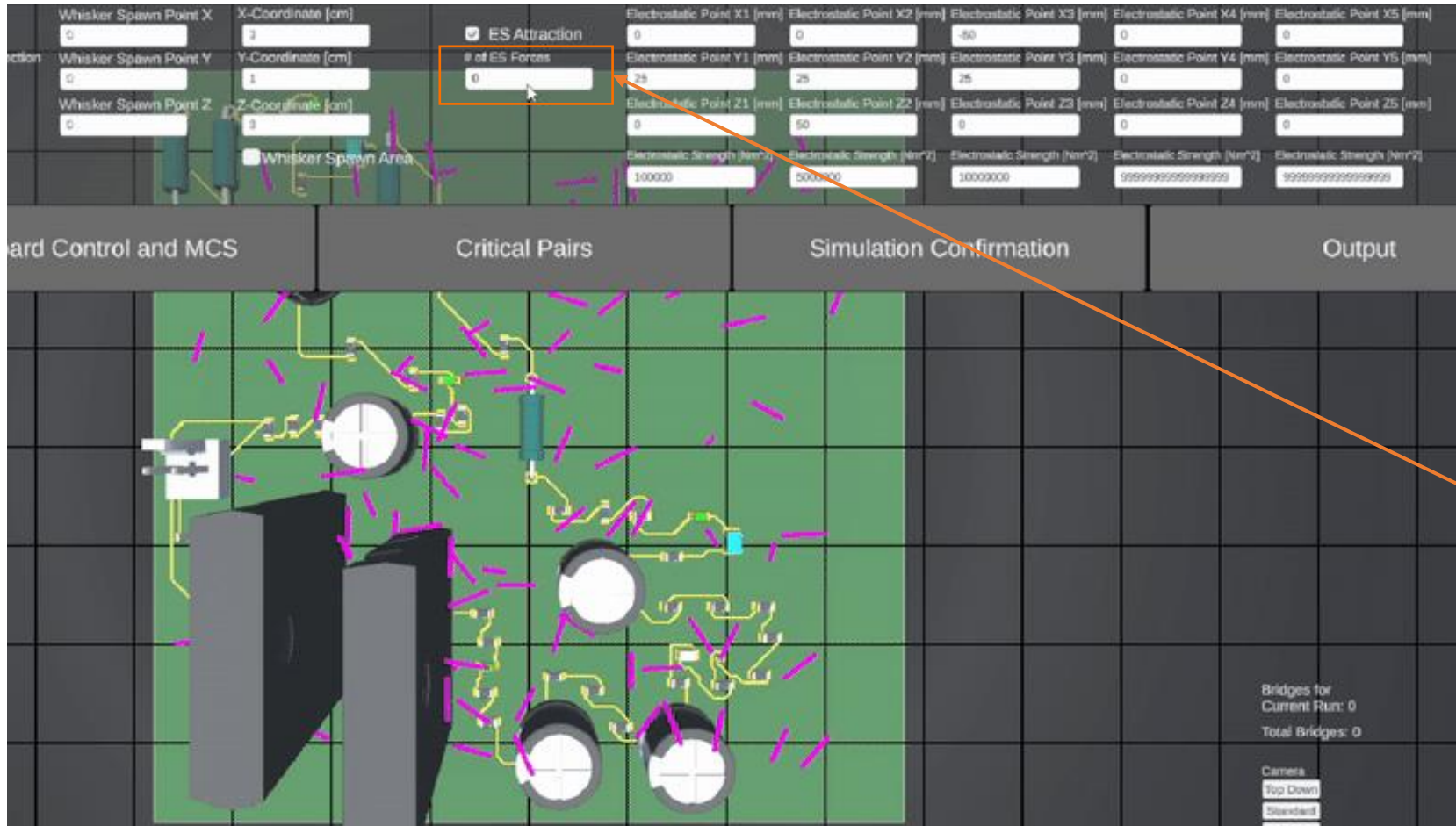
Phase Three of
ES Attraction Development:
Multiple ES Forces

of ES Forces (0-5)

Shown: As # of ES
increases from 1 to 2 to 3,
the whiskers are pulled
between ESP1, ESP2, and
ESP3 (respectively).

ESP4 and ESP5 do not have
an effect since # of ES does
not reach 4 or 5.

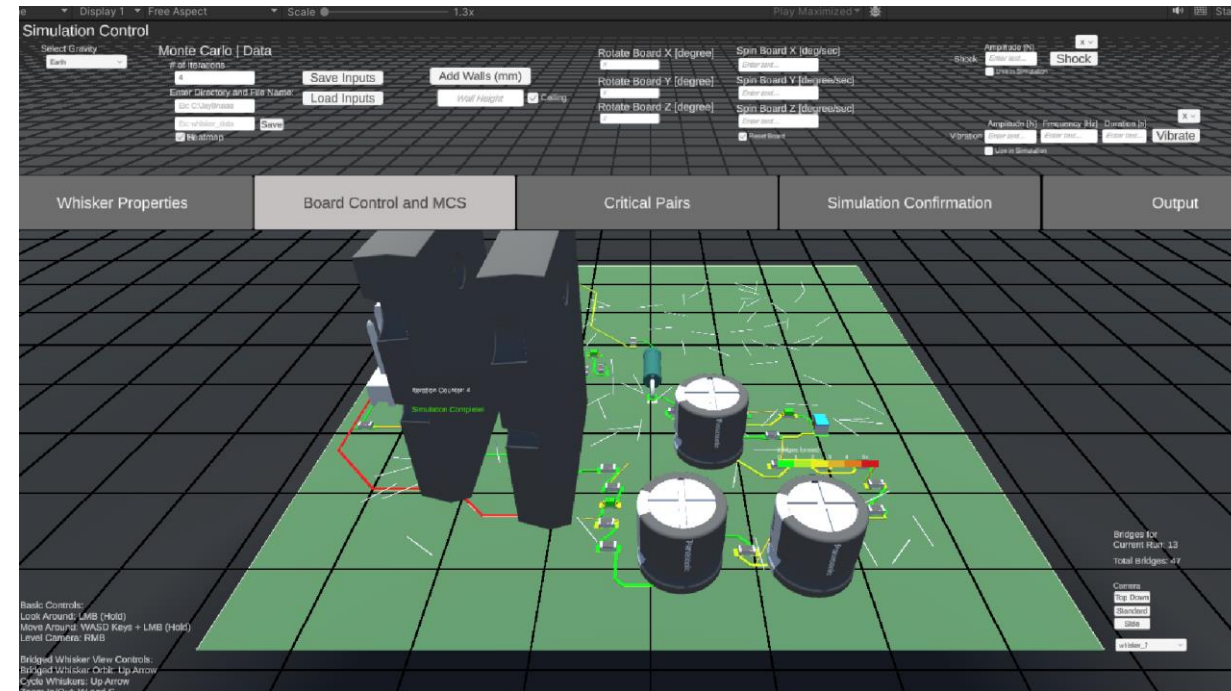
GIF



Development, Testing, and Results

Heatmap – Intro

- Wanted to change the current heatmap to visually color the entire PCB instead of tagging conductors
- Use color gradients to show intensity levels
- Overlay the heatmap on the UI using the original button



Development, Testing, and Results

Heatmap – Testing

- Rewrote script to generate a “Texture2D” heatmap
- Mapped trigger positions to texture coordinates on a PCB plane
- Applied the texture to material on PCB and toggled it with the button

```
Texture2D tex = new Texture2D(textureRes, textureRes);  
...  
colors[index] = ping >= 5 ? Color.red : ...;  
tex.SetPixels32(colors);  
tex.Apply();  
pcbMaterial.mainTexture = tex;
```


Development, Testing, and Results

Heatmap – Discussion

- Variable names overlapped but didn't match exactly
- Original script had hidden references in the hierarchy
- Original script wasn't built for texture-based heatmaps
- Trigger positions had to be converted into texture-space
- Unity inspector references made debugging difficult

Development, Testing, and Results

Simulation Profile

Vibration and Shock Timeline Development

Vibration	Enter text...	Enter text...	Enter text...	Enter text...
	Amplitude [N]	Frequency [Hz]	Duration [s]	Vibration Start [s]
Shock	Enter text...	Enter text...		
	Amplitude [N]	Shock Start [s] 1/2 Period of Sine[s]		

Vibration Input Added:
Time Start

Shock Input Added:
 $\frac{1}{2}$ Sine Period Input to
calculate Time Duration
and Frequency, Time Start

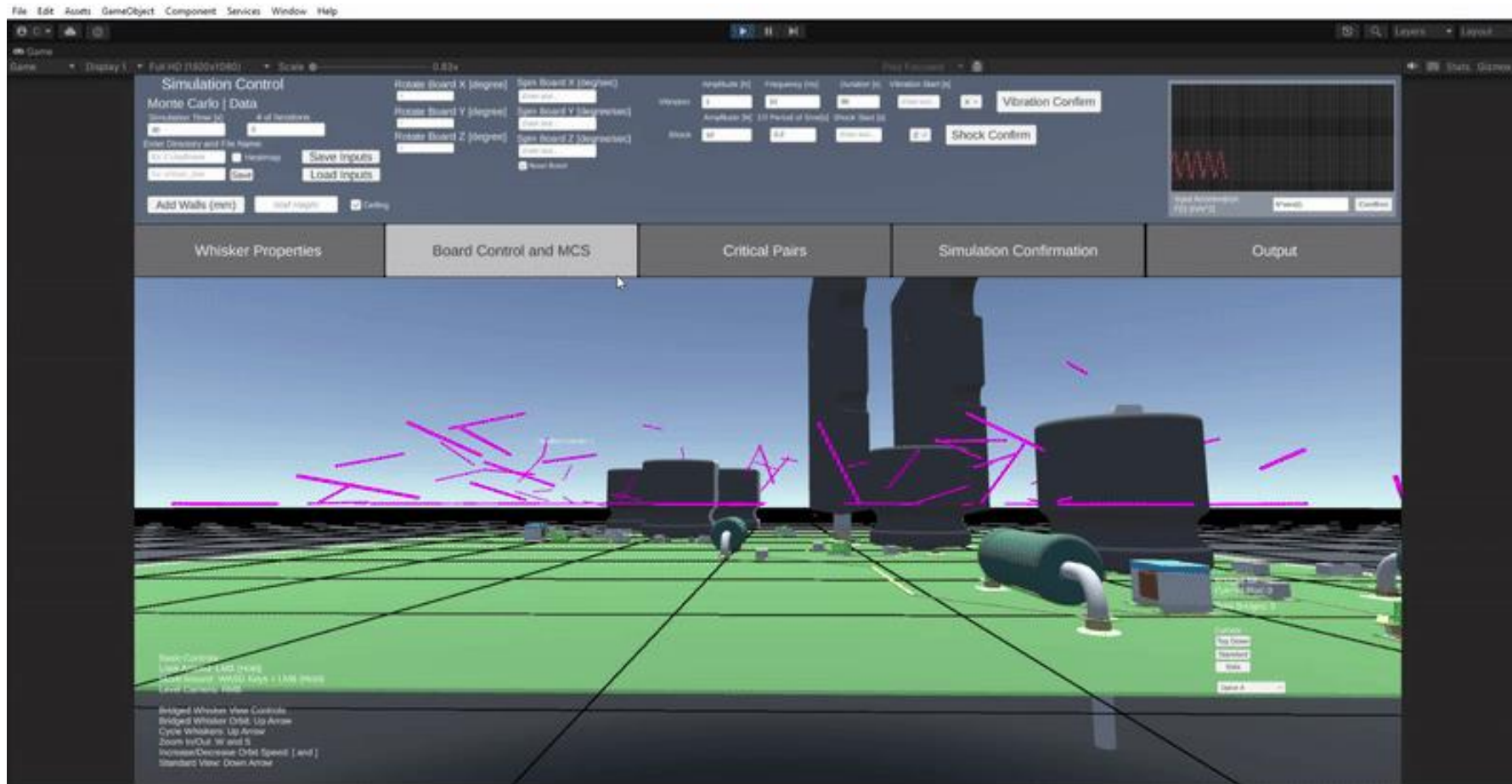
GIF

Development, Testing, and Results

Simulation Profile

Time-dependent
Acceleration Development

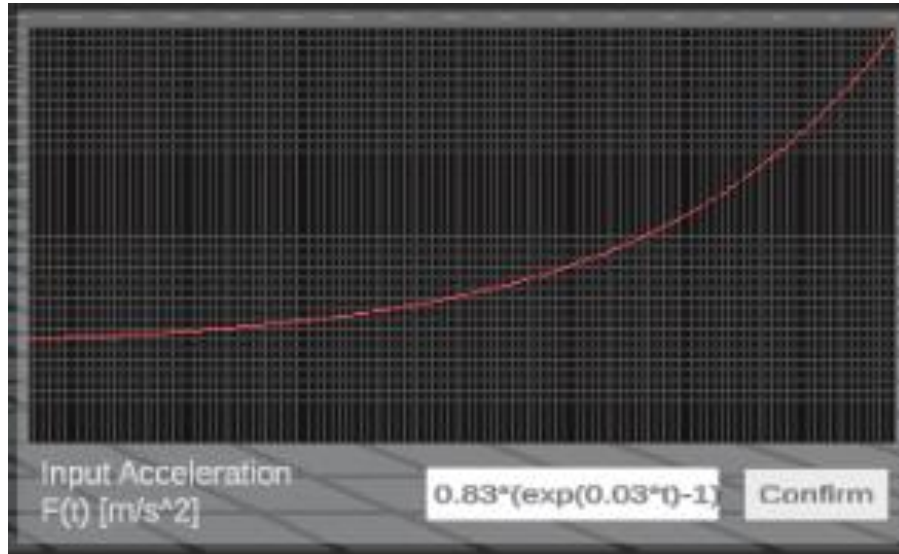
Acceleration Input Added:
Time Length, Acceleration
Function



GIF

Development, Testing, and Results

Simulation Profile



Simulation Time [s]
120

Profile Development

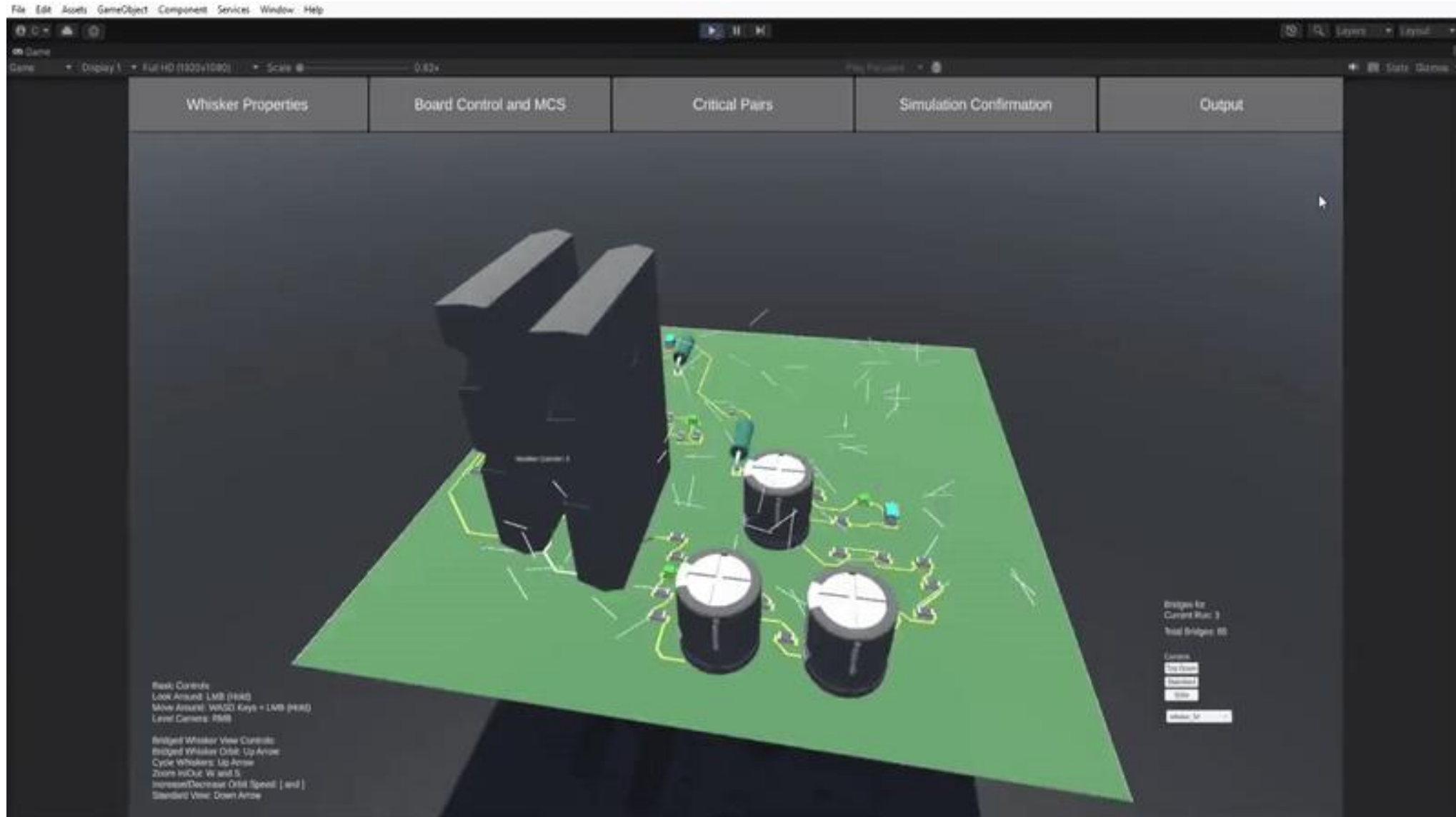
Saving/Loading of Monte Carlo Inputs

Default Setting for Simulation Time, Acceleration Input, Shock Inputs, and Vibration Inputs

	Amplitude [N]	Frequency [Hz]	Duration [s]	Vibration Start [s]	
Vibration	1	10	90	10	X v Vibration Confirm
	Amplitude [N]	1/2 Period of Sine[s]	Shock Start [s]		
Shock	10	0.2	110	Z v Shock Confirm	

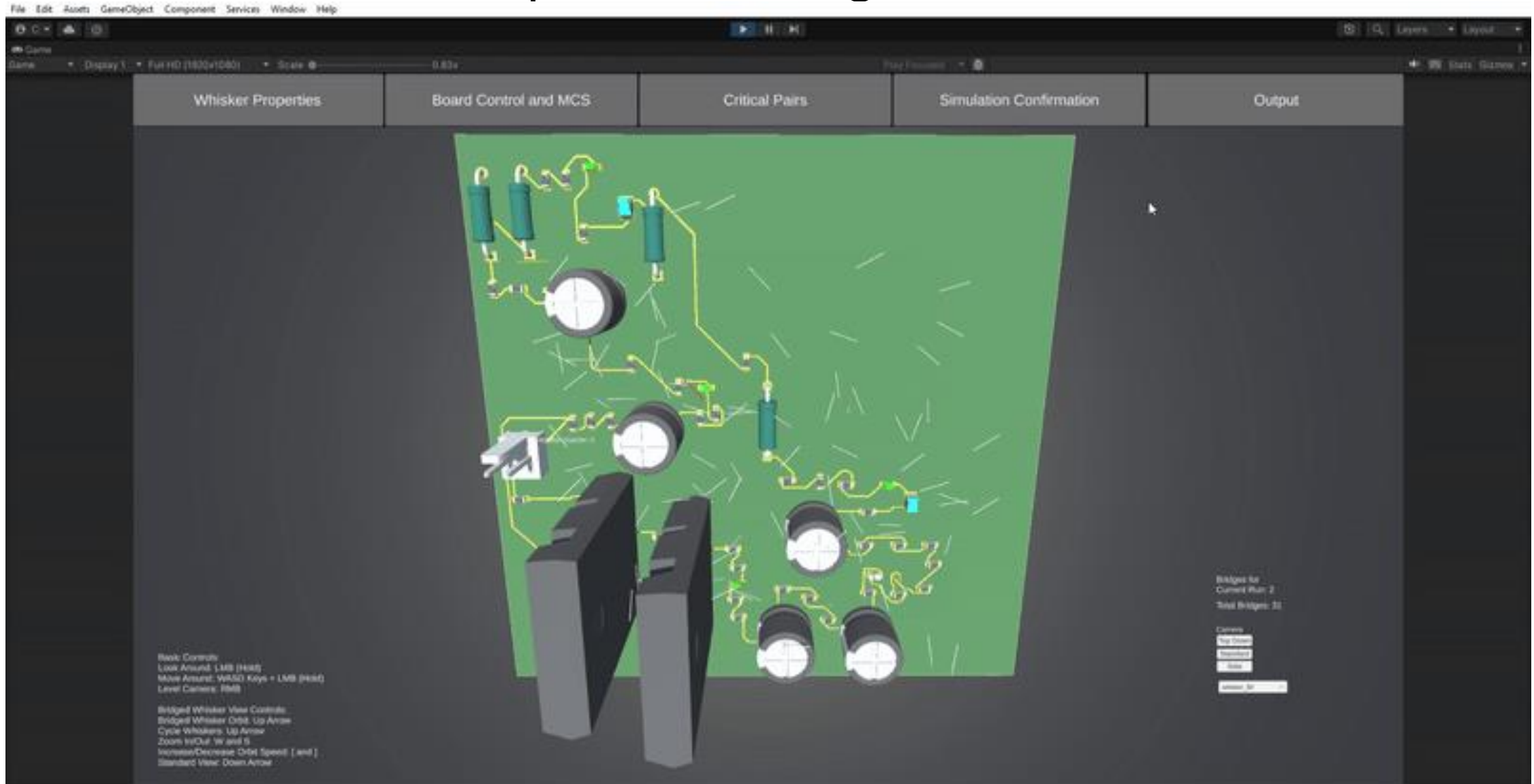
GIF

Development, Testing, and Results



GIF

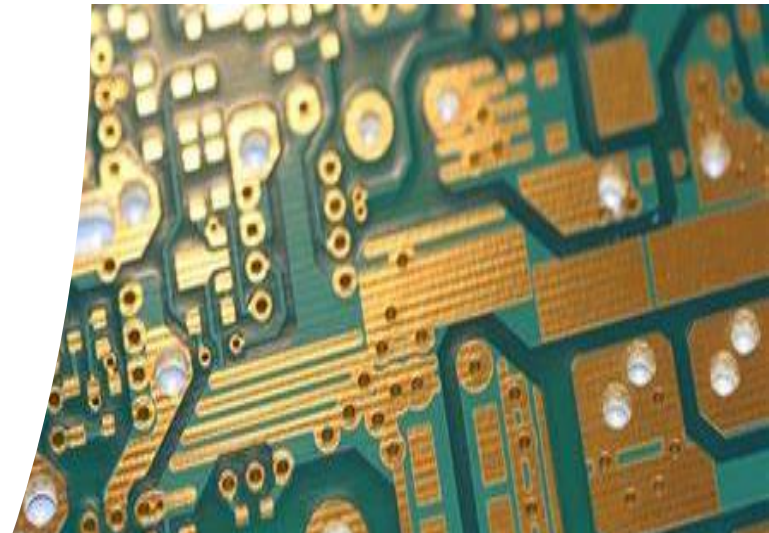
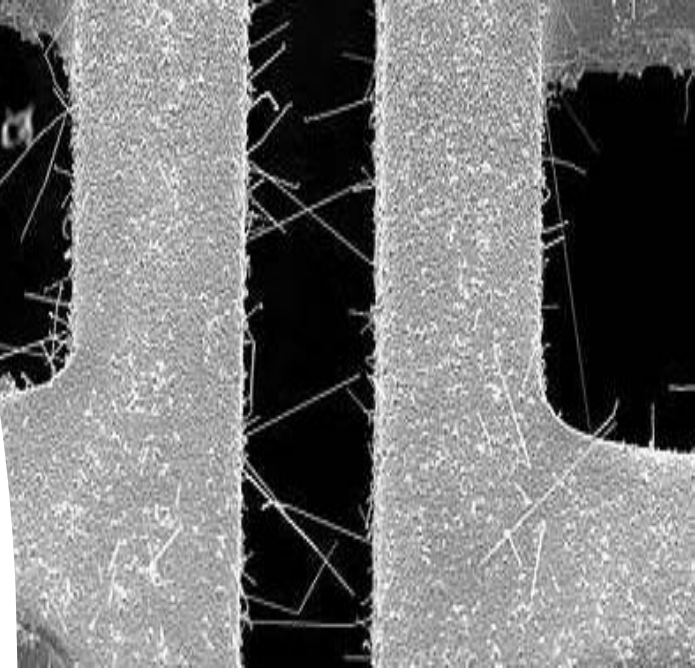
Development, Testing, and Results



GIF

Conclusion

- Project Goal
 - Develop a more capable tool for whisker simulation analysis, identifying PCB design weaknesses.
- Achievements
 - Added external force functionality to the simulation for more varied scenarios.
 - Further developed the simulation save/load functionality, setting for standardized testing.
 - Testing and development into the uploading of various PCBs.
- Obstacles
 - Unity 3D is not primarily a data simulation tool, requiring development in all areas.
 - Handoff from previous efforts sometimes does not provide all the information.
 - Inexperience in Unity software and C# code scripting provided difficulties.



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- [7] SpaceX, “Falcon User’s Guide”, Sept. 2021, Available: [Falcon Payload User's Guide](#)